The Influence of Market Sentiment on Commercial Bank Stability

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Abstract

We investigate how market-based uncertainty and sentiment shocks transmit to the systemic connectedness of major US financial institutions across distinct states of risk appetite. Using daily data spanning the post-2022 period, we combine equity implied volatility (VIX), gold, and crude oil uncertainty measures, along with cryptocurrency sentiment, within a Quantile-VAR connectedness framework that measures spillovers across pessimistic, typical, and exuberant market states. The methodology delivers state-dependent estimates of average connectedness, net directional spillovers, and network roles, and it further evaluates joint channels by pairing VIX with non-equity uncertainty/sentiment measures. Results reveal strong regime dependence. Under lower-tail conditions, gold and oil uncertainty act as net shock absorbers, whereas VIX and cryptocurrency sentiment exhibit comparatively weak standalone effects; when combined with VIX, non-equity channels typically exhibit stronger stabilizing capacity. In median conditions, uncertainty measures are predominantly stabilizing, consistent with shock absorption in tranquil markets. In upper-tail states, VIX becomes a dominant transmitter of systemic instability, whereas cryptocurrency sentiment is associated with enhanced shock absorption, implying that speculative activity in digital-asset markets can, under certain conditions, reduce pressure on bank interconnectedness.

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JEL Classification: E52; E58; G11; G12; G15.

1. Introduction

##add our research?

Despite the growth of fintech and neobanking structures, traditional banking institutions remain central nodes in modern financial intermediation, transforming maturity and liquidity, providing payment and market-making services, and concentrating exposures that can transmit shocks to the broader economy [Paroush, 1988, Boyd et al., 2005, Laeven, 2011, Anginer et al., 2014]. Historical episodes, ranging from the Global Financial Crisis to more recent banking stress, underscore that banking disruptions can propagate rapidly through market prices, funding conditions, and confidence [Calomiris and Jaremski, 2024]. A core implication for research and policy is that systemic outcomes are not well captured by institution-by-institution assessments; they emerge from the structure and intensity of linkages among large intermediaries and the markets in which they are priced [Anginer et al., 2014].

The post-2020 environment has further sharpened this challenge. Russia's invasion of Ukraine triggered sharp shifts in geopolitical risk, commodity prices, and inflation expectations, with spillovers into market volatility and financial conditions. Central banks' subsequent tightening cycle, together with intermittent risk-off episodes, heightened the sensitivity of bank equity valuations to shocks that originate outside the banking sector (for example, energy-market turmoil, broad equity uncertainty, and changes in risk appetite). In this setting, understanding when and through which channels uncertainty becomes systemically relevant is as important as measuring its average effects.

A long tradition in finance emphasizes that risk appetite and beliefs shape asset prices and cross-market reallocation, especially in extremes. Investor sentiment can move valuations in ways that are difficult to arbitrage and are therefore most visible when uncertainty is elevated or when narratives dominate fundamentals. Closely related behavioral mechanisms, shifts in confidence, and speculative belief dynamics, provide a useful organizing framework for why markets may overshoot in downturns and reprice risk aggressively in booms [Shiller, 2014].

Market-based uncertainty measures provide an operational way to quantify these mechanisms in real time. Yet the risk environment relevant to banks is multi-dimensional, where investors reallocate not only across equities, but also towards traditional hedging venues such as gold and commodities, and increasingly towards digital-asset markets. Gold is often found to act as a hedge and, in some episodes, a safe haven, although that property is not invariant across shocks or periods [Salisu et al., 2021, Xu et al., 2024]. Digital-asset markets are more difficult to classify. Recent work on stablecoins and Bitcoin suggests that these markets are closely intertwined with broader monetary and risk conditions rather than standing outside them [Karau, 2023, Hoang and Baur, 2024].

A second motivation is methodological. Standard connectedness and spillover measures based on forecast-error variance decompositions in VAR systems are informative about shock transmission, but they typically summarize dynamics around the conditional mean. By construction, such approaches can obscure the asymmetries that matter most in financial markets, where contagion can intensify in crashes but also strengthen in exuberant rallies, and the identity of net transmitters can change across states. Quantile-based connectedness methods address this limitation by allowing the transmission mechanism to differ across the return distribution, thereby aligning inference directly with pessimistic, normal, and exuberant market conditions [Chatziantoniou et al., 2021].

Against this background, the paper studies state-dependent connectedness between US systemically important financial institutions and a set of uncertainty and sentiment measures spanning traditional and digital asset markets. The contribution is twofold. First, it links the systemic-risk literature on banking fragility to the literature on uncertainty and bank behavior [Buch et al., 2015, Demir and Danisman, 2021, Wu et al., 2022]. Second, it embeds digital-asset sentiment within the same empirical architecture as conventional uncertainty measures, allowing a direct comparison of stabilizing and destabilizing roles across regimes [Karau, 2023, Hoang and Baur, 2024].

The results show pronounced regime dependence in banking-system connectedness. In pessimistic (lower-tail) states, gold and oil uncertainty act as net absorbers of systemic risk, while equity implied volatility and cryptocurrency sentiment exhibit comparatively weak standalone effects; combining VIX with other uncertainty measures tends to strengthen the stabilizing role of the non-equity channel. In typical (median) conditions, uncertainty measures are largely stabilizing for the bank network, consistent with risk being absorbed rather than amplified when markets are not in the tails. In exuberant (upper-tail) states, equity implied volatility becomes a dominant net transmitter of systemic instability, whereas cryptocurrency sentiment is associated with stronger shock absorption, suggesting that speculative demand in digital-asset markets can, at times, reduce pressure on bank interconnectedness, while institution-level spillover leadership shifts across banks in a manner consistent with state-contingent amplification mechanisms.

The rest of this paper proceeds as follows: Section 2 provides a concise overview of the key research that provides a foundation for this research's theoretical and empirical focus. Section 3 presents a clear oversight of the data used in this analysis, along with a thorough explanation of the Q-VAR and TCI methodologies utilised. Section 4 presents the results of the analyses conducted, while Section 5 provides a thorough discussion surrounding this work's policy and regulatory implications, along with several directions for future research. Finally, Section 6 concludes.

2. Previous Literature

This research is related to several areas of prior literature on the determinants of bank stock market performance and risk. The first stream focuses on the effects of policy and wider decision-making on the banking sector. Central banking actions have been identified to directly mitigate systemic risk [Matousek et al., 2026] while the Basel Committee regulatory framework strengthens bank transparency and the resilience of the financial system [Mies, 2024]. Expansionary monetary policy, in particular, has been identified to lower bank default risk [Soenen and Vander Vennet, 2022, Arghyrou et al., 2024], while low market capital and high regulatory capital make the international risk-taking channel of monetary policy more effective [Avdjiev and Serena, 2025]. Considering fiscal actions and banking relationships, government responses to economic crises and bank defaults have increased systemic risk, for example, during periods of exceptional risk such as the pandemic [Duan et al., 2021]. Moreover, fiscal problems translating into higher sovereign debt risks have been found to suppress bank performance and increase bank stock volatility, implying a deterioration of the risk-return relation [Liu et al., 2025].

The second area of the literature focuses on the relationship between the banking system and the broader economy. Examining the size and impact of banks shows that those not large enough to be characterized as '*Too-Big-To-Fail*' together exert greater contagious pressures on the real economy and exhibit similar behavior [Li and Lai, 2024]. Furthermore, banks with global exposures, especially systemically important and less capitalized banks, are sources of US systemic risk [McLemore et al., 2022]. Regarding the bail-in–bail-out dilemma during crises, bail-ins have the potential to reduce systemic risk but are less effective at preventing systemic crises [Siebenbrunner et al., 2024]; however, bail-outs also destabilize the system, as higher lending by bailed-out banks increases non-performing loans [Sivá, 2025].

The third stream of research related to this paper concerns the connection between uncertainty measures and systemic (in)stability. The economic policy uncertainty is negatively associated with bank tail risk [Fang et al., 2023], but positively related to systemic risk [Wu et al., 2022, Fang et al., 2023]. Moreover, EPU affects the monetary policy-bank risk relationship by weakening the impact of the 'risk-taking' channel of monetary policy [Wu et al., 2022]. Geopolitical uncertainty is influential on bank performance [Boungou and Urom, 2025], while broadly destabilizing the banking system [Shabir et al., 2023]. When considering climate policy uncertainty, it has been found to lower bank systemic risk, especially in banking institutions with a larger systemic effect and in highly competitive banking sectors [Liu et al., 2024], and to induce lower loan loss provisions [Zhang et al., 2026].

By centering interest on market uncertainties, policy news is a strong source of instability

towards financial markets [Conrad et al., 2025, Baker et al., 2026], also shaping investor sentiment, which affects market stability and efficiency [Zhou, 2025]. The VIX constitutes the most reliable forecasting tool for systemic risk in developed economies, especially during crises [Feng et al., 2026], exerting strong effects on all equity sectors [Kang et al., 2023]. While the stock market generates instability that influences the wider financial system [Xu et al., 2024], the VIX exhibits a higher ability to forecast in the short-run in bear markets, while in the long-run in bull markets [Li and Ma, 2025]. Furthermore, stock markets are closely linked to oil markets [Bagirov and Mateus, 2026], with oil shocks (mostly negative) having a substantial impact on stock index volatility [Cadena-Silva et al., 2025]. Gold markets serve as systemic stabilizers against shocks from stocks [Xu et al., 2024, Hanif et al., 2025], but this ability diminishes as volatility increases in gold markets [Faraj et al., 2025]. Previous literature confirms a research gap in understanding how uncertainties arising from various asset types affect the systemic resilience of the globally influential US banking system. Particular emphasis is required regarding tail-risk behaviors driven by investor irrationality.

3. Data and Methodology Employed

This study uses daily observations for four market-sentiment proxies and the market values of eight large US financial institutions. The sentiment variables are the CBOE Volatility Index (VIX), which proxies equity-market fear; the CBOE Gold ETF Volatility Index (GoldV), which captures expected variability in gold markets; the CBOE Crude Oil ETF Volatility Index (OilV), which measures expected variability in crude-oil markets; and the Crypto Fear & Greed Index (CryptoS), which summarizes sentiment in cryptocurrency markets. VIX, GoldV, and OilV are obtained from the Federal Reserve Economic Data database, while CryptoS is collected from alternative.me. The banking block comprises JP Morgan Chase, Bank of America, Citigroup, Wells Fargo, Goldman Sachs, Morgan Stanley, US Bancorp, and Visa, with daily market-value series retrieved from Datastream. The sample spans 24 February 2022 to 8 January 2026, a period that covers the Russia-Ukraine war, the associated inflationary episode, the 2023 banking turmoil, and recurrent bouts of financial-market stress.

Insert Figure 1 about here

Figure 1 presents evidence of the clear heterogeneity across the four examined uncertainty proxies. VIX and OilV are prone to intermittent periods of sharp elevation, while GoldV is the most contained of the conventional uncertainty measures, fluctuating within a narrower band and rising only gradually towards the end of the sample. By contrast, CryptoS is visibly the most erratic

series, displaying the largest volatility and the most pronounced regime shifts over time. It is clear that market sentiment is not synchronised across asset classes, where cryptocurrency sentiment is the most unstable component, whereas gold-market uncertainty is the most contained. Further, figure 2 presents evidence that the share-price trajectories of the selected institutions are ultimately upward over the sample, but the strength of that revaluation differs markedly across firms.

Insert Figure 2 about here

Table 1 confirms that the transformed series are highly heterogeneous and non-Gaussian. Among the uncertainty indices, CryptoS has by far the largest variance (202.958), whereas GoldV has the smallest (23.273), indicating a far more unstable profile for cryptocurrency sentiment than for the conventional uncertainty measures. Among the examined financial institutions, Goldman Sachs has the highest mean daily log-difference (0.107), followed by JP Morgan Chase (0.087), while US Bancorp has the lowest mean (0.004) and the highest variance in the financial institution subsample (3.822), implying the weakest mean-variance profile in the group. In addition, all series exhibit excess kurtosis, the Jarque-Bera statistics reject normality, the ERS statistics support stationarity after transformation, and the $Q^2(10)$ results indicate pervasive nonlinearity and volatility clustering. These properties justify the use of a quantile-based framework rather than an empirical framework centered only on conditional means.

Insert Table 1 about here

As some of the selected variables are drawn from different data providers, the sample is aligned on common observation dates before estimation. We let $X_{m,t}$ denote the level of variable m on day t. All series are utilised as continuously compounded first differences, defined as:

$$r_{m,t} = 100\left[\ln(X_{m,t}) - \ln(X_{m,t-1})\right]. \tag{1}$$

Utilising log-differences serves two purposes. First, it places the selected volatility indices, the cryptocurrency-sentiment index, and financial institutions market-value series on a common scale of daily percentage changes. Second, it removes persistent level effects that would otherwise contaminate the dynamic spillover analysis. The transformation is applied uniformly across all variables so that the connectedness measures are interpretable as transmission of shocks in daily changes rather than in raw levels. The empirical structure is then estimated across several configurations. In the single-sentiment specification, the endogenous vector stacks one sentiment proxy and the

eight bank series, so that $k = 9$. In the paired-sentiment specifications, the vector includes two sentiment proxies and the eight banks, so that $k = 10$. The state vector can therefore be written generically as $y_t = (s_{1,t}, \ldots, s_{q,t}, b_{1,t}, \ldots, b_{8,t})'$, where $q \in \{1, 2\}$, $s_{1,t}$ and $s_{2,t}$ denote the relevant sentiment measures, while $b_{n,t}$ denotes the transformed market-value series of bank n.[1]

The statistical properties of the transformed data further motivate a quantile-based approach, as presented in Table 1, indicating marked heterogeneity in dispersion across sentiment measures, with crypto sentiment exhibiting the largest variability and gold volatility the smallest. The transformed series are also distinctly non-Gaussian, exhibiting skewness and excess kurtosis, whereas the unit-root diagnostics reported indicate that the log-difference transformation is sufficient to address non-stationarity concerns. These features matter for identification because a mean-based model can obscure behavior in the tails of the distribution, precisely where systemic fragility is most likely to emerge. For that reason, the selected analysis focuses on the lower, median, and upper conditional quantiles of the joint distribution, which are interpreted as stress, normal, and exuberant market states, respectively.

To quantify how sentiment shocks and financial institution stock price valuations interact across the entire conditional distribution, we employ the quantile vector autoregression connectedness framework developed in Chatziantoniou et al. [2021] and extended in Cunado et al. [2023]. The benefit of selecting this empirical framework is that it allows the propagation of shocks to differ between adverse, typical, and exuberant market states, where the lower-tail quantile $\tau = 0.05$ is interpreted as a stress or bear regime, $\tau = 0.50$ as the central regime, and $\tau = 0.95$ as an exuberant or bull regime. Because volatility indices and financial institution market-value changes are skewed, leptokurtic, and potentially nonlinear, modeling only the conditional mean would obscure the tail behavior central to systemic-risk analysis.

Therefore, for a given system of dimension k, let y_t denote the $k \times 1$ vector of endogenous variables observed on day t. At quantile $\tau \in (0, 1)$, the QVAR(p) is written as

$$y_t = \mu(\tau) + \sum_{\ell=1}^{p} \Phi_\ell(\tau) y_{t-\ell} + u_t(\tau). \tag{2}$$

In this expression, $\mu(\tau)$ is a $k \times 1$ vector of quantile-specific intercepts, $\Phi_\ell(\tau)$ is the $k \times k$ matrix of autoregressive slope coefficients at lag ℓ, and $u_t(\tau)$ is the vector of quantile residuals. The coefficients are quantile-specific, so the dependence structure is allowed to change across the lower, central, and upper sections of the distribution rather than being forced to remain constant. Each

[1] This notation is useful because the econometric design is identical across all model variants; only the composition of the sentiment block changes.

equation within the system is estimated by quantile regression. Let $y_{i,t}$ denote the ith element of y_t and define the regressor vector $x_t = (1, y'_{t-1}, \ldots, y'_{t-p})'$. The parameter vector for equation i at quantile τ is obtained from

$$\beta\widehat{\beta}_i(\tau) = \arg\min_{\beta_i} \sum_{t=p+1}^{T} \rho_\tau(y_{i,t} - x'_t\beta_i)\,, \qquad \rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\}). \tag{3}$$

where $\rho_\tau(\cdot)$ asymmetrically weights positive and negative residuals, which is the key reason quantile regression can recover heterogeneous dependence in the tails. In the current setting, this feature is essential because shocks during episodes of extreme pessimism or exuberance need not propagate in the same way as shocks under normal market conditions. Under the usual stability condition, the QVAR admits the moving-average representation

$$y_t = \mu(\tau) + \sum_{h=0}^{\infty} A_h(\tau) u_{t-h}(\tau), \tag{4}$$

with

$$A_0(\tau) = I_k, \qquad A_h(\tau) = \sum_{\ell=1}^{\min(h,p)} \Phi_\ell(\tau) A_{h-\ell}(\tau), \quad h \geq 1. \tag{5}$$

The matrices $A_h(\tau)$ map an innovation at time $t-h$ into the current value of the system at horizon h. They therefore summarize the quantile-specific propagation mechanism and form the basis for the forecast-error variance decompositions used below.

Connectedness is derived from the generalized forecast error variance decomposition of Koop et al. [1996] and Pesaran and Shin [1998], adapted to the quantile setting. For a forecast horizon H, the contribution of shocks in variable j to the H-step-ahead forecast error variance of variable i at quantile τ is given by

$$\theta^H_{i,j}(\tau) = \frac{\Sigma^{-1}_{jj}(\tau) \sum_{h=0}^{H-1} \left[e'_i A_h(\tau)\Sigma(\tau)e_j\right]^2}{\sum_{h=0}^{H-1} e'_i A_h(\tau)\Sigma(\tau)A'_h(\tau)e_i}. \tag{6}$$

Here, $\Sigma(\tau)$ is the covariance matrix of the quantile residuals, $\Sigma_{jj}(\tau)$ is its jth diagonal element, and e_i and e_j are $k \times 1$ selection vectors with unity in positions i and j, respectively, and zeros elsewhere. The numerator isolates the effect of shocks originating in variable j on variable i, while the denominator scales that contribution by the total forecast error variance of variable i. Because the generalized decomposition does not orthogonalize shocks, the resulting spillovers are invariant to variable ordering, which is particularly desirable when no strong prior exists regarding the causal

ranking between sentiment and bank variables. The raw generalized shares do not necessarily sum to one across columns, so they are normalized as

$$\theta\widetilde{\theta}^{H}_{i,j}(\tau) = \frac{\theta^{H}_{i,j}(\tau)}{\sum_{m=1}^{k} \theta^{H}_{i,m}(\tau)}. \tag{7}$$

After normalization, $\theta\widetilde{\theta}^{H}_{i,j}(\tau)$ can be interpreted as the fraction of variable i's H-step forecast error variance that is attributable to innovations in variable j, conditional on the quantile under study. This normalization is necessary for coherent aggregation into node-level and system-level connectedness measures.

Node-level spillovers are constructed from the normalized variance shares. The total spillovers transmitted by variable i to the rest of the system are

$$C^{H}_{i\to\bullet}(\tau) = \sum_{\substack{j=1\\ j\neq i}}^{k} \theta\widetilde{\theta}^{H}_{j,i}(\tau), \tag{8}$$

whereas the spillovers received by variable i from all other variables are:

$$C^{H}_{\bullet\to i}(\tau) = \sum_{\substack{j=1\\ j\neq i}}^{k} \theta\widetilde{\theta}^{H}_{i,j}(\tau). \tag{9}$$

The distinction between the two measures is substantive. A large $C^{H}_{i\to\bullet}(\tau)$ means that shocks originating in variable i explain a sizable share of the forecast uncertainty of the remaining variables, so i acts as a shock transmitter. A large $C^{H}_{\bullet\to i}(\tau)$ means that variable i is primarily a shock receiver. When i corresponds to a sentiment proxy, and the remaining nodes correspond to banks, the two measures indicate whether that sentiment variable destabilizes the banking block or instead absorbs shocks from it. The net spillover position of variable i is defined as:

$$C^{H,\mathrm{NET}}_{i}(\tau) = C^{H}_{i\to\bullet}(\tau) - C^{H}_{\bullet\to i}(\tau). \tag{10}$$

Positive values indicate net transmitters of shocks, whereas negative values indicate net absorbers. This is the quantity used to interpret the stabilizing or destabilizing role of each sentiment proxy and each bank in the network graphs and in the extended joint connectedness analysis. System-wide fragility is summarized by the total connectedness index,

$$TCI^H(\tau) = \frac{1}{k}\sum_{i=1}^{k} C^H_{\bullet\to i}(\tau) = \frac{1}{k}\sum_{i=1}^{k}\sum_{\substack{j=1\\ j\neq i}}^{k} \theta\widetilde{\theta}^H_{i,j}(\tau). \tag{11}$$

The TCI measures the average share of forecast error variance attributable to cross-variable spillovers rather than to own shocks. Values closer to zero indicate weak interdependence and low systemic fragility, whereas values closer to one indicate dense cross-market transmission and a greater probability that shocks propagate through the network. In this study, a higher TCI indicates that market sentiment and bank market values are tightly coupled, so idiosyncratic disturbances can more easily become system-wide.

The connectedness measures are computed separately for $\tau = 0.05$, $\tau = 0.50$, and $\tau = 0.95$, which correspond to the lower, central, and upper regions of the conditional distribution. This design allows the same set of variables to exhibit different transmission patterns under stress, under ordinary conditions, and under exuberant market sentiment. For each model specification, the QVAR is estimated using a 100-day rolling window, and the variance decompositions are evaluated at a 10-day forecast horizon. Rolling estimation is crucial because the transmission mechanism is not assumed to be constant over events, such as the Russia-Ukraine geopolitical crisis and the subsequent inflationary period that followed thereafter; instead, the intensity and direction of spillovers are allowed to evolve with changing market conditions. Three complementary outputs are then obtained from the rolling normalized GFEVDs. First, the averaged dynamic connectedness measures report the mean spillover structure across specifications and quantiles. Second, the net extended joint connectedness measures track over time whether a sentiment proxy or a bank is a net transmitter or a net receiver of shocks once the relevant system has been specified. Third, the network-connectedness representation maps the same information onto a directed network, where nodes represent sentiment proxies and banks, and link intensities are weighted by the corresponding normalized spillover shares. Taken together, these statistics provide a coherent picture of how stock, gold, oil, and cryptocurrency sentiment shape the resilience of the US banking system across markedly different market states.

4. Empirical Results

When considering extreme pessimism as represented by the 5th percentile, Table 2 shows that neither the VIX nor cryptocurrency volatility (CryptoS) exerts a significant net impact on banking-sector systemic risk. In contrast, gold volatility (GoldV) and oil volatility (OilV) each serve as modest stabilizers, absorbing risk during downturns. When combined with the VIX, the role of

these assets is accentuated: the VIX–Gold pairing makes gold's stabilizing effect substantially stronger, whereas the VIX–Oil combination still yields only moderate stability. Similarly, when VIX is paired with CryptoS, cryptocurrency volatility becomes a weak-to-moderate stabilizer of the system. On the institutional side, Wells Fargo emerges as the dominant source of systemic risk under these stress conditions (when OilV and CryptoS are included). However, including the VIX in the mix shifts some of that risk away: Wells Fargo's relative impact diminishes and Morgan Stanley and JPMorgan Chase become co-primary contributors to systemic risk. In other words, during severe market stress conventional fear indices and crypto uncertainty have little one-sided effect on fragility, whereas traditional "safe-haven" assets (gold and oil) absorb shocks. This pattern reflects a flight-to-quality dynamic: under panic, investors flock to safe assets that dampen systemic stress, and elevated market volatility tends to redistribute risk across banks rather than amplifying it in one institution.

Insert Table 2 about here

Under typical market conditions as presented in Table 3, all measured uncertainties act as strong net absorbers of systemic instability. Individually, the volatilities of the stock market (VIX), gold, oil, and cryptocurrency each reduce systemic risk, and the effect is particularly pronounced for Visa (a proxy for normal banking conditions). In joint settings, the non-VIX asset dominates the interaction: for example, pairing VIX with gold or with oil or with CryptoS makes that commodity or crypto volatility an even stronger stabilizer, while VIX's own stabilizing influence remains only weak to moderate. Morgan Stanley and Citigroup consistently emerge as contributors to systemic risk in this scenario, but only modestly so. In sum, during normal times the banking system appears relatively resilient: uncertainty measures (and diversifying holdings) tend to soak up volatility rather than transmit it. Banks like Morgan Stanley and Citigroup are not major destabilizers, suggesting broad diversification and healthy capital buffers in ordinary conditions.

Insert Table 3 about here

During extreme optimism, as represented when examining the 95th percentile, evidence as presented in Table 4 suggests that the system behaves inherently differently. In this situation, the VIX becomes a strong source of systemic instability, with elevated stock-market volatility under exuberance driving banking risk upward. GoldV and OilV continue to play stabilizing roles (now at moderate strength), while CryptoS shows a strong stabilizing effect. Notably, the weakest bank

in our sample, US Bancorp, consistently emerges as the primary source of systemic risk under these conditions. These findings suggest that in exuberant periods, elevated stock market volatility and the lure of "lucrative but safe" investments shape systemic risk, where investors' search for yield (or perceived safety) in alternative assets actually mitigates stress in the banking sector. Diversifying assets, including digital and fintech instruments, is found to act as a stabilizer when the blue-chip market is particularly volatile. Intriguingly, our results imply a counterintuitive stabilizing role for irrational cryptocurrency speculation, where heightened CryptoS (reflecting exuberant trading) appears to temper bank risk by overemphasizing market signals. This extends the classic flight-to-quality theory, where even under optimism, agents seek out safe assets, and the presence of high-risk "safe-haven" assets (such as cryptocurrency and fintech options) can enhance stability by redirecting speculative flows away from banks. In effect, both chronically under-performing assets (for example, weak banks) and highly volatile speculative assets increase investors' incentives to seek out the most stable stores of value, reinforcing a "rationally speculative" flight-to-quality dynamic (as in the pseudo-wealth framework of Guzman and Stiglitz [2021]).

Insert Table 4 about here

Each set of quantile results, therefore, uncovers a consistent theme that flight-to-quality behavior dominates system dynamics. During episodes of financial market deterioration, specifically episodes of market crashes, traditional safe havens (gold, oil) are found to absorb shocks; whereas, during phases of substantial growth, even speculative assets play a safety-valve role. This differential behavior of risks across quantiles underscores why both conservative and speculative assets can jointly stabilize the financial system under different market states.

4.1. Net pairwise directional connectedness

###RUN PARALLEL ANALYSIS HERE TO VERIFY THESE POINTS

Next, we examine the time-varying net extended joint connectedness measures at the lower (5th), middle (50th), and upper (95th) quantiles of the return distributions. These plots identify whether each uncertainty/sentiment measure acts, on net, as (i) a transmitter of spillovers to the banking system (destabilizer) or (ii) a receiver/absorber of spillovers from the system (stabilizer). Throughout, values above (below) the zero line indicate net transmission (net absorption) of connectedness.

Insert Figure 3 about here

Figure 3 reports the net connectedness of each uncertainty/sentiment measure when considered separately alongside the banking group under bear-market conditions ($\tau = 0.05$). The VIX emerges as an episodic, but important net transmitter of spillovers to bank equities. This effect is most evident in the initial phase of the Russia–Ukraine war (2022) and re-intensifies during late 2024 and early 2025. Gold-market volatility (GoldV) is most influential at the onset of the conflict, but thereafter it predominantly acts as a net absorber of systemic risk, consistent with a stabilizing role once the initial repricing associated with the war subsides. By contrast, cryptocurrency sentiment (CryptoS) is the strongest net source of instability for the banking system, with the net spillover to banks peaking in late 2024 and early 2025. Importantly, CryptoS exhibits pronounced state dependence: while it is a net transmitter during some stress episodes, it is more often a stabilizer during the 2023 banking turmoil and during the later phase of heightened geopolitical risk (2025).[2]

To clarify how spillovers are modified when uncertainty channels interact, Figure 3 reports pairwise systems that include the VIX, a second uncertainty proxy, and the banking sector. It shows that when GoldV is included in the system, the VIX becomes a stronger net transmitter of spillovers to banks, suggesting that stock-market volatility has a larger destabilizing footprint when gold markets are simultaneously volatile. Conversely, conditioning on the VIX modestly strengthens GoldV's net absorbing role, indicating that gold volatility (and the associated hedging channel) provides slightly stronger system-wide protection when equity volatility is elevated. Adding oil-market volatility (OilV) partly mutes the VIX's net influence on the banking sector; symmetrically, including the VIX weakens OilV's alternating (dual) role. Finally, the results indicate that when CryptoS is included, cryptocurrencies absorb part of the instability generated by the VIX. In the opposite direction, including the VIX slightly reduces CryptoS's hedging capacity and does not offset the pronounced destabilizing spillovers originating from CryptoS in early 2025.

Insert Figure 4 about here

At the median quantile (Figure 4), representing normal-market conditions ($\tau = 0.50$), the uncertainty/sentiment measures are predominantly net absorbers of connectedness, implying that they mostly mitigate, rather than amplify, banking-sector instability under typical market conditions.

[2]These episodes broadly coincide with three identifiable shock clusters within the sample: the Russian invasion of Ukraine on 24 February 2022; the March-May 2023 US regional-banking turmoil, including the failures of Silicon Valley Bank (10 March 2023), Signature Bank (12 March 2023), and First Republic Bank (1 May 2023); and the late-2024/early-2025 return of elevated geopolitical and market uncertainty, marked by escalation in the Russia-Ukraine conflict in November 2024 and a sharp post-election repricing in digital assets, with Bitcoin surging towards and above US$100,000 before renewed volatility around the January 2025 US presidential inauguration. These events are included to contextualise the timing of the spillover spikes rather than to claim direct causal identification.

The VIX is the weakest stabilizer in this state, while OilV and CryptoS provide the strongest net absorption. Focusing on the pairwise specifications, the Figure 4 indicates that the net role of the VIX is largely unchanged when GoldV is introduced, whereas GoldV becomes slightly more effective at absorbing systemic risk when the VIX is included in the system. For the VIX–OilV pairing, both variables modestly reduce the hedging capacity of the other, suggesting partial overlap in their stabilizing channels. Finally, the VIX is largely unaffected by the inclusion of CryptoS, whereas CryptoS becomes a somewhat stronger stabilizer when stock-market volatility is explicitly accounted for.

Insert Figure 5 about here

At the upper quantile (Figure 5) representing bull-market conditions ($\tau = 0.95$), net spillovers alternate in sign across time, highlighting the dual role of uncertainty measures during periods of extreme positive banking returns. The VIX is the dominant net transmitter of spillovers and therefore the main source of systemic instability in this state. GoldV also becomes a major transmitter, and in several episodes, its destabilizing effect exceeds that of CryptoS. Focusing on the pairwise systems, results suggest that GoldV has a mildly moderating effect on the VIX on average; however, the interaction is time-varying. In the early phase of the Russia-Ukraine conflict, the presence of the VIX is associated with GoldV acting as a strong destabilizer, whereas in later phases, the VIX appears to mute GoldV's spillovers. The VIX–OilV pair indicates weak but symmetric attenuation, that OilV slightly reduces the VIX's destabilizing footprint, and the VIX similarly dampens OilV's net spillover effects.

4.2. Network connectedness

We next examine the network connectedness structure to identify the principal transmitters and receivers of spillovers among the banks and the uncertainty/sentiment measures across market states. Figure 6 (VIX–banks) indicates that, under the denoted bear-market conditions ($\tau = 0.05$), most banks, along with the VIX, act as net transmitters of connectedness, implying broad-based propagation of shocks within the banking system. At the same time, the largest institutions (for example, JP Morgan Chase and Bank of America) act as net absorbers, contributing disproportionately to the system's resilience in the lower tail. Figure 7 extends the analysis to other uncertainty proxies. GoldV, OilV, and CryptoS primarily behave as net absorbers of systemic risk, in contrast to the more mixed (dual) role observed for the VIX. The VIX–OilV combination strengthens system resilience, although Goldman Sachs remains a major source of net spillovers. In the cases of GoldV and CryptoS, JP Morgan Chase and Wells Fargo emerge as key destabilizing forces.

Insert Figures 6 & 7 about here

When considering the denoted normal-market conditions ($\tau = 0.50$), in Figure 8, the VIX and Visa are the only consistent stabilizers under normal conditions, absorbing spillovers generated by the remaining institutions. Introducing the remaining uncertainties in pairs with the VIX adds further stabilizers and reinforces the existing ones. This indicates that, during calm periods, stock-market volatility does not amplify banking instability and that other asset-market uncertainty channels similarly fail to threaten the banking system's resilience. The near-universal net transmission originating from banks further suggests that, absent crisis or euphoria, systemic interactions are not dominated by a single "leader" bank that drives herding behavior.

Insert Figures 8 & 9 about here

In Figure 9 under bull-market conditions ($\tau = 0.95$), stock-market volatility and the top-performing bank (Goldman Sachs) are found to be the main net transmitters of systemic risk, while lower-cap banks, poor performers (US Bancorp), and Visa are net absorbers that stabilize the system. Considering paired uncertainties, GoldV efficiently mitigates the destabilizing effect of the VIX, while OilV and CryptoS also provide effective hedges against VIX-driven spillovers. Leadership patterns differ across market channels, where the top performers exhibit higher systemic risk when gold volatility is included, whereas large-cap banks play a leading role when oil volatility is considered. Notably, under bull-market banking conditions, crypto sentiment does not destabilize bank markets, suggesting that strong performance in traditional financial institutions is accompanied by insulation from shocks originating in modern digital-asset markets.

Fundamentally, stock-market volatility emerges as the most consistent threat to systemic stability in the banking sector, corroborating the VIX's major systemic role in Feng et al. [2026]. Across specifications and market states, gold is the most reliable stabilizer, followed by crude oil and cryptocurrency sentiment. This pattern implies that the potential reshaping of large-cap, traditional, and systemically important institutions is more tightly linked to conventional financial-market forces than to innovative alternative investments. This stands in contrast to the view that digital currencies constitute the dominant source of instability for the established banking system, as in Podhorsky [2024]. Gold's role as a persistent resilience mechanism aligns with its safe-haven character, consistent with Hanif et al. [2025] and Salisu et al. [2021]. Finally, evidence that high-performing banks and stock-market exuberance are associated with heightened instability supports an interpretation in which risk appetite is not confined to assets with a high propensity

to form asset-pricing bubbles. During crises, the flight-to-safety mechanism manifests as allocations favoring safe-haven profit-makers (gold) alongside speculative cryptocurrency exposure, with weaker-performing bank equities also acting as stabilizers. This combination of safe-asset demand and performance-seeking behavior is consistent with a "rationally speculative" strategy mix and provides an empirically grounded channel through which the pseudo-wealth mechanism of Guzman and Stiglitz [2021] can be reflected in market dynamics.

5. Associated Discussion, Policy Implications and Directions for Future Research

This research presents a central contribution by showing that systemic fragility in the United States banking equity network is strongly state-dependent, where market-based connectedness is not a background condition that drifts smoothly over time but a regime feature that can jump abruptly when sentiment moves into extremes. This is particularly important for modern macro-finance discussions because the sample spans a period in which macro fundamentals and market mood were repeatedly forced into contact—beginning with the escalation of the Russia–Ukraine war on February 24, 2022, and continuing through subsequent volatility episodes that included acute stress in parts of the US banking sector in 2023.

The results present evidence of a pronounced U-shape in systemic connectedness across market states, where TCI is high in both lower-tail (pessimistic) and upper-tail (optimistic) regimes, but materially lower in median conditions. In practical terms, the banking network behaves as a loosely coupled system under normal conditions, but becomes tightly synchronized, almost mechanically prone to amplification, when the system enters either extreme fear or extreme greed. Furthermore, this pattern is sufficiently large, where in the lower tail, TCI values cluster around roughly 89–92%; in the median state, they fall into the mid-60s (roughly 65–69%); and in the upper tail, they again rise to around 89–93% depending on the system specification. Therefore, systemic risk is not maximized only in “bad times” (crashes) alone, but also during euphoric conditions, when optimism compresses perceived risk and encourages synchronized positioning. A natural explanation for the identified U-shape draws on the idea that extreme sentiment compresses heterogeneity. When investors’ narratives become unusually confident or unusually fearful, decision rules converge—herding becomes more likely, and idiosyncratic shocks are less likely to remain idiosyncratic. This aligns with the Keynesian notion that investment activity depends heavily on spontaneous optimism, and with modern behavioral accounts in which contagious narratives generate correlated shifts in beliefs and risk taking. In connectedness terms, tails look like high common factor states, where shocks that would be absorbed locally in normal times transmit broadly because balance sheets and investor portfolios are being repriced against the same dominant narrative.

A second result of particular interest is that the VIX plays a regime-specific leadership role. The VIX is designed to measure 30-day expected volatility implied by S&P 500 option prices, and it is widely treated as a barometer of market-implied fear and uncertainty precisely because option prices embed the market's willingness to pay for insurance. The results indicate that not only does the VIX exert influence, but also that its direction of influence depends strongly on the system's position in the conditional distribution. When periods of optimism are considered, the VIX is a major net transmitter of shocks to the banking system, whereas at the median it is a net absorber (and hence stabilizing). This is consistent across several system specifications, including those that pair the VIX with other sentiment indices, where VIX remains a net positive contributor in the upper tail, whereas other indices frequently become net absorbers. Fundamentally, such high-risk-appetite states are not safe simply because asset prices are rising; they can be fragile precisely because the system's pricing of risk is compressed and therefore sensitive to volatility repricing. A potential explanatory mechanism is through a leverage–risk-constraint channel. When risk appetite is elevated, intermediaries and the investors who hold their equity often behave as if effective risk capacity has expanded; leverage and exposure can rise endogenously during periods of rising market conditions (pro-cyclical leverage), and risk constraints (VaR-style or margin-based) become binding when measuring volatility elevation. In such environments, an increase in implied equity volatility is a balance-sheet event, as it tightens risk limits, increases margin requirements, and can generate forced deleveraging and liquidity spirals—dynamics that are theoretically emphasized in models linking funding liquidity, margins, and market liquidity. The finding that the VIX becomes a dominant shock transmitter in the optimistic tail is therefore interpretable as the market's risk-on equilibrium, being vulnerable to abrupt volatility repricing: a volatility shock converts a high-confidence state into a joint sell/hedge episode, producing an elevation of connectedness.

When considering safe-haven and cross-asset hedging channels, in particular, the median state, the full set of uncertainty proxies, VIX, gold volatility, crude oil volatility, and cryptocurrency sentiment, tend to be net absorbers rather than net transmitters, implying that in normal conditions these markets (and the trading strategies they represent) mute rather than amplify banking shocks. A particularly interesting result is identified when considering the tail states, that even when banking connectedness is high, some non-bank sentiment proxies act as net stabilizers, and they often become more stabilizing when VIX is included in the system. Gold's stabilizing role in the broader literature is well established but conditional, where gold can behave as a hedge and, at times, a safe haven in crisis-like periods, though its effectiveness can vary by horizon and episode. It is important to note that GVZ is an option-implied price of insurance in gold markets, an estimate of expected 30-day volatility constructed with VIX-style methodology applied to options on the SPDR Gold

Shares ETF (GLD). When such an index is a net absorber of banking shocks, such gold-derivative markets provide a liquid, scalable venue for volatility/uncertainty expression that substitutes for dumping banking exposures. Put differently, part of the system's demand for convexity and protection is diverted into the gold options complex rather than being expressed entirely through bank equity repricing. However, the crude-oil volatility proxy (OVX) can be read analogously, as it is an estimate of the expected 30-day volatility of crude oil as priced by options on the United States Oil Fund (USO). While oil is not regarded as a safe haven in the same sense as gold, oil volatility is a macro-uncertainty object (geopolitics, inflation, growth) and is widely embedded in hedging and cross-asset trading strategies. Results indicating that oil volatility frequently absorbs net shocks, particularly when paired with VIX in some specifications, suggest that commodity-volatility markets can also serve as shock absorbers by supplying hedging capacity when equity volatility risk is repriced. Further, when VIX enters the system jointly with GVZ/OVX/crypto sentiment, the indices often become stronger net stabilizers, suggesting an integrated hedging network, where investors do not hedge equity-volatility risk in one market only; instead, they reallocate risk across a menu of volatility and sentiment instruments. Such behavior is consistent with broader evidence that implied-volatility measures and cross-asset spillovers are deeply intertwined in practice, with VIX often exerting a strong influence over equity sectors and interacting with other hedging assets. Such a network can provide resilience only as long as the hedging venues remain liquid and operational under stress.

5.1. Policy Implications

One of the key novel results concerns cryptocurrency sentiment, which is found to be a net stabilizer of the banking system across several regimes, including optimistic tail states. Since this index summarizes cryptocurrency market fear and greed, it can be read as a proxy for the intensity of speculative attention and risk-on emotion in a parallel speculative arena. It can be interpreted that cryptocurrency markets may, in some episodes, function as a speculative overflow reservoir. When optimism in traditional risky assets is high, marginal speculative capital can be redirected into cryptocurrencies in search of convex payoffs, narrative-driven returns, or social-momentum trades. In such a case, part of the aggregate demand for speculative exposure is reallocated away from banking equities, muting the amplification of bank-specific shocks. Such an interpretation, however, cannot be treated as a structural guarantee. When considering cryptocurrency as a potential stabilizing mechanism, this must be considered within the context of regime-conditional and market-structure-dependent, where cryptocurrency sentiment can absorb speculative heat when it remains a substitute arena for risk taking, but it can become a fault line if the banking system

becomes directly exposed to cryptocurrency runs, stablecoin stress, or correlated deleveraging in cryptocurrency-derivative markets.

The findings motivate a policy stance that is explicitly regime-aware, in which system risk is not well summarized by a single mid-cycle metric, as connectedness can jump by 20–25 percentage points between median- and tail-regime regimes. Considering this, four implications follow. First, systemic monitoring should explicitly integrate sentiment-conditioned market indicators into routine monitoring. Secondly, stress testing should consider volatility reversals from high-risk-appetite starting points, not only recession from neutral baselines.[3] In practical terms, the goal is not to replace recession scenarios, but to recognize that the banking system can be most interconnected precisely when euphoria makes it look safest. Thirdly, capital buffer design should be interpreted through both credit cycles and volatility cycles. The Basel III countercyclical capital buffer (CCyB) is explicitly motivated by the desire to protect the banking sector from the build-up of system-wide risk associated with excess aggregate credit growth and the macro-financial environment. The presented results suggest that an additional macroprudential signal, market-implied volatility during optimistic states, may be informative about when the system is vulnerable to contagion even absent a conventional credit boom signal. Finally, resilience depends on the functioning of cross-asset hedging venues, including commodities and (increasingly) cryptocurrency markets. Results indicating that gold- and oil-volatility indices and cryptocurrency sentiment often absorb net shocks, especially when equity volatility is present, imply that these markets can serve as pressure valves for the banking system.

5.2. Directions for Future Research

The results open avenues for several directions for future research. One is to connect state-dependent spillovers to bank financial strength and funding fragility. This research shows that spillover roles migrate across regimes; a natural extension is to clarify whether these migrations are systematically related to capital strength, liquidity buffers, and funding composition. A second direction is to separate sentiment-driven volatility shocks from information about bank fundamentals. Because VIX and other implied-volatility measures embed forward-looking insurance prices, future work can investigate when periods of volatility elevation primarily reflect macro news and when they represent self-reinforcing risk repricing. A third direction is to study cross-regime in-

[3] The Fed's supervisory stress tests are designed to ensure large banks can continue lending under severe recession scenarios and include narratives with abrupt declines in risk appetite, impaired market functioning, and heightened volatility. The presented evidence adds a specific refinement, that even in optimistic states, a volatility shock can generate extremely high connectedness (and thus rapid spillover transmission), implying that a "good-times-to-volatility-shock" transition may be a distinct and relevant macroprudential scenario class.

teractions across asset classes. This research finds evidence of complementarity, where adding VIX often strengthens stabilizers, suggesting that "mismatched regimes" (for example, equity calm with cryptocurrency fear, or commodity stress with equity optimism) may be the states in which hedging networks either perform best or fail. A deeper mapping of regime mismatches would sharpen practical inferences about when diversification channels are reliable and when they collapse into common shocks. Fourth, one could decompose the financial network into layers, such as commercial banks, broker-dealer-like market intermediaries, and payment/fintech nodes, and investigate whether some layers systematically absorb shocks in the tails.

One could also include macro-financial feedback more explicitly into the interpretation. Stress-test narratives and financial stability reports emphasize that risk appetite can fall abruptly and market functioning can be impaired; however, the presented results suggest that the banking network may be most spillover-prone precisely when risk appetite is high, and volatility repricing arrives as a surprise. Future research can therefore clarify how monetary and financial conditions interact with the sentiment–connectedness channel, shaping when volatility becomes a banking-system destabilizer. Finally, one could specifically investigate the global synchronization of volatility regimes and their implications for US bank stability. International volatility benchmarks, such as Europe's VSTOXX, are option-implied measures analogous in spirit to the VIX and reflect expected volatility in Eurozone equities. When global equity-volatility regimes shift together, spillovers may become non-additive, where synchronized volatility repricing could tighten the network more than separate regional shocks would imply. This direction is also motivated by evidence that the 2023 banking turmoil exhibited cross-border contagion in banking-sector linkages, even though interdependence evolved afterward.

6. Concluding Comments

This study examines how market-implied uncertainty and sentiment shocks are transmitted to (and absorbed by) the equity valuations of systemically important US financial institutions across distinct states of market risk appetite. We utilise a Q-VAR connectedness framework with extended joint and network connectedness measures, allowing spillovers to differ systematically between stress (lower-tail), typical (median), and exuberant (upper-tail) regimes, while focusing on equity-market volatility (VIX), option-implied gold and crude-oil volatility measures, and cryptocurrency market sentiment. These choices ensure that the estimated spillover network is anchored in forward-looking, market-priced uncertainty as well as a parallel sentiment channel originating in digital-asset markets.

Results indicate that systemic fragility in the banking equity network is strongly state dependent. Total connectedness is materially elevated in both pessimistic and euphoric regimes, while it

declines in median conditions, indicating a pronounced U-shaped relationship between the tightness of the banking network and the distributional state of returns. This result implies that systemic vulnerability is not confined to downturns; it also increases during periods of exceptional optimism, when correlated positioning and rapid risk repricing can synchronize institutions and amplify common shocks. Consequently, assessments that rely on average conditions can understate system-wide spillover risk precisely when the market environment is most prone to transition into or out of extreme sentiment states.

Second, the evidence identifies equity-market volatility as the most regime-sensitive and economically important driver of spillover transmission. In exuberant states, the VIX becomes a dominant net transmitter of shocks to the banking system, consistent with the interpretation that volatility repricing acts as a balance-sheet and risk-constraint event that tightens funding and margin conditions and accelerates the propagation of shocks across bank equities. In the lower tail, the VIX exhibits a more episodic and comparatively muted net role, implying that extreme stress does not mechanically translate into a uniform volatility-driven contagion channel; rather, leadership and amplification depend on which uncertainty channels are simultaneously active. This asymmetry underscores the importance of modeling tail states directly, as the sign and magnitude of the volatility channel cannot be reliably inferred from mean-based dynamics.

A further conclusion concerns the stabilizing capacity of non-equity uncertainty channels and their interaction with the VIX. Gold-related uncertainty is the most reliable absorber of banking-sector shocks across regimes, whereas crude-oil uncertainty plays a more mixed role, with stabilizing capacity that depends on the prevailing state and on whether equity volatility is jointly priced. Importantly, when equity volatility is considered alongside these alternative channels, the non-equity indices often become more stabilizing on net, consistent with the existence of an integrated hedging network in which investors reallocate protection demand across several liquid volatility and sentiment classes rather than expressing it solely through bank-equity repricing, emphasizing that systemic resilience is partly shaped by the depth and functioning of cross-asset hedging markets, especially when volatility conditions shift quickly.

Results also provide a particularly nuanced finding with regard to cryptocurrency market sentiment. Across several specifications, particularly in normal and optimistic states, cryptocurrency sentiment behaves as a net absorber of banking-sector spillovers, suggesting that speculative intensity in digital-asset markets can, at times, act as a pressure valve that draws marginal risk-taking away from bank equities. At the same time, the estimates indicate pronounced state dependence, where cryptocurrency sentiment can serve as a transmission channel during specific stress episodes, implying that the stabilizing role is not structural and may weaken as interlinkages between banks

and cryptocurrency markets deepen. Framed in theoretical terms, the results are consistent with a hybrid mechanism in which flight-to-quality behavior (toward robust hedges such as gold) coexists with a sentiment-driven reallocation of speculative demand across asset classes—an empirical pattern that aligns with "rationally speculative" portfolio behavior and is compatible with pseudo-wealth interpretations in which perceived wealth and risk appetite shift with sentiment rather than fundamentals alone.

Such conclusions yield direct implications for systemic financial market oversight and surveillance, along with direct implications for stress-test design. For investors and bank risk managers, it is important to note that diversification and hedging effectiveness are regime contingent. In median conditions, uncertainty channels often mitigate banking spillovers, while in tail regimes—especially under exuberance—equity-volatility shocks can dominate and compress the benefits of naive diversification within the banking sector. Cross-asset hedges associated with gold (and, more conditionally, oil) demonstrate more consistent stabilizing capacity, implying that hedging strategies should be designed as state-aware portfolios that treat volatility repricing risk as a central driver of systemic spillovers.

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Table 1: Descriptive Statistics of Selected Analysed Variables (logarithmic differences)

	Mean	Variance	Skewness	Ex.Kurtosis	JB	ERS	Q(10)	Q2(10)
				Uncertainty indices				
VIX	-0.079	53.814	1.050***	8.846***	3,348.033***	-4.548***	8.515	105.532***
GoldV	0.031	23.273	1.144***	4.535***	1,045.039***	-10.950***	10.391*	141.374***
OilV	-0.031	30.169	0.416***	3.238***	452.802***	-13.328***	19.629***	133.459***
CryptoS	0.027	202.958	-0.244***	8.974***	3,271.292***	-9.555***	29.244***	130.384***
				Major banks				
JP Morgan Chase	0.087	2.443	0.001	5.485***	1,216.355***	-4.046***	4.501	44.859***
Bank of America	0.029	2.987	-0.179**	4.264***	741.484***	-3.379***	10.520*	66.730***
Citigroup	0.073	3.451	0.008	6.123***	1,518.426***	-5.148***	6.03	47.232***
Wells Fargo	0.064	3.548	-0.016	4.211***	718.372***	-4.128***	8.971	28.811***
Goldman Sachs	0.107	3.041	0.212***	5.819***	1,378.802***	-3.690***	5.571	63.956***
Morgan Stanley	0.075	3.237	0.073	4.727***	905.714***	-3.807***	8.046	61.534***
US Bancorp	0.004	3.822	-0.273***	3.864***	616.823***	-3.144***	2.098	174.757***
Visa	0.051	1.86	-0.162**	4.360***	773.974***	-7.938***	7.988	83.533***

Note: This table reports descriptive and diagnostic statistics for the analysed uncertainty indices and banking institutions. All series are expressed in logged form. Reported statistics include the mean, variance, skewness, excess kurtosis, minimum and maximum values, the Jarque–Bera (JB) test for normality, the Elliott–Rothenberg–Stock (ERS) unit-root test, and the Ljung–Box Q(10) and $Q^2(10)$ statistics for linear and non-linear dependence, respectively. The results indicate substantial heterogeneity across variables, pervasive departures from normality (skewness and excess kurtosis), and strong evidence of stationarity across all series. Moreover, the Q(10) and $Q^2(10)$ statistics reveal significant serial correlation and volatility clustering, highlighting the importance of accounting for higher-order dependence and time-varying volatility in subsequent empirical analysis.

Table 2: Estimated Averaged Dynamic Connectedness at the 5th Quantile

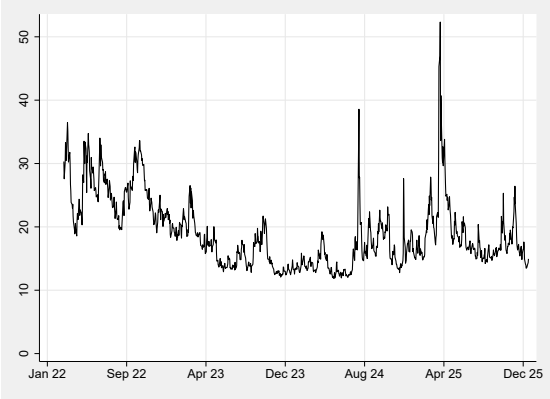
	VIX	IndexV	JPM	BoAm	Citi	W.Far	Gold.S	Morg.S	US Banc	Visa	FROM
						VIX and major banks					
VIX	24.57	-	9.28	9.19	9.46	9.58	9.33	9.55	9.64	9.39	75.43
JP Morgan	9.64	-	6.82	12.16	12.31	11.91	12.11	12.22	11.66	11.18	93.18
Bank of America	9.39	-	12.13	5.23	12.52	12.42	12.42	12.42	12.39	11.07	94.77
Citigroup	9.18	-	11.90	12.13	8.06	11.89	12.10	11.80	11.97	10.98	91.94
Wells Fargo	9.34	-	11.67	12.10	11.99	8.38	11.75	11.77	12.02	10.97	91.62
Goldman Sachs	9.30	-	11.69	11.86	12.08	11.92	7.95	12.49	11.82	10.89	92.04
Morgan Stanley	9.56	-	11.82	11.96	12.02	12.02	12.47	7.06	11.89	11.19	92.94
US Bancorp	10.33	-	11.61	11.85	12.03	12.02	11.66	11.83	7.66	11.01	92.34
Visa	10.11	-	11.38	11.36	11.72	11.25	11.80	11.75	11.50	9.13	90.87
TO	76.85	-	91.49	92.81	94.13	93.00	93.65	93.84	92.89	86.68	
NET	1.43	-	-1.70	-2.16	2.18	1.39	1.61	0.90	0.55	-4.20	TCI=90.57
						GoldV and major banks					
GoldV	-	24.45	9.67	9.49	9.30	9.47	9.03	9.42	9.70	9.48	75.55
JP Morgan	-	8.67	9.10	12.49	12.15	11.74	11.27	11.97	11.28	11.34	90.90
Bank of America	-	8.31	12.48	6.69	12.49	12.83	11.80	12.37	12.13	10.91	93.31
Citigroup	-	8.15	12.10	12.68	8.94	12.21	11.48	11.82	11.71	10.91	91.06
Wells Fargo	-	8.08	11.59	12.45	11.99	10.59	11.20	11.62	11.89	10.58	89.41
Goldman Sachs	-	8.26	11.95	12.25	12.11	11.74	9.14	12.38	11.47	10.69	90.86
Morgan Stanley	-	8.48	12.17	12.48	12.17	12.08	11.88	7.90	11.69	11.13	92.10
US Bancorp	-	8.61	11.44	12.39	11.89	12.44	11.01	11.37	10.24	10.61	89.76
Visa	-	9.12	11.52	11.51	11.58	11.66	10.77	11.41	11.46	10.96	89.04
TO		67.70	92.91	95.74	93.68	94.17	88.45	92.35	91.33	85.64	
NET		-7.85	2.01	2.44	2.62	4.75	-2.41	0.26	1.58	-3.39	TCI=89.11
						VIX, GoldV and major banks					
VIX	14.97	11.12	9.17	8.92	9.24	9.49	9.36	8.29	8.99	9.44	85.03
GoldV	11.95	11.43	9.54	9.21	9.68	9.62	9.58	9.73	9.70	9.56	88.57
JP Morgan	9.38	8.48	7.06	10.76	11.07	10.67	10.99	11.05	10.16	10.40	92.95
Bank of America	8.82	8.28	11.16	4.62	11.52	11.49	11.43	11.40	10.99	10.29	95.38
Citigroup	8.71	8.15	10.94	11.06	7.11	10.98	11.02	11.01	10.69	10.34	92.89
Wells Fargo	8.98	8.14	10.55	10.73	10.82	8.71	10.77	10.73	10.50	10.07	91.29
Goldman Sachs	9.30	8.45	10.66	10.61	10.87	10.83	7.19	11.30	10.64	10.15	92.81
Morgan Stanley	9.06	8.48	10.83	10.67	10.99	10.94	11.35	6.88	10.54	10.24	93.12
US Bancorp	9.61	8.56	10.69	10.72	10.95	11.09	10.65	10.80	6.68	10.26	93.32
Visa	9.51	8.63	10.48	10.26	10.83	10.50	10.68	10.75	10.19	8.07	91.93
TO	85.43	78.27	94.03	92.94	95.96	95.61	95.83	96.07	92.41	90.75	
NET	0.40	-10.30	1.07	-2.44	3.07	4.32	3.02	2.95	-0.92	-1.18	TCI=91.73

(continued on the next page)

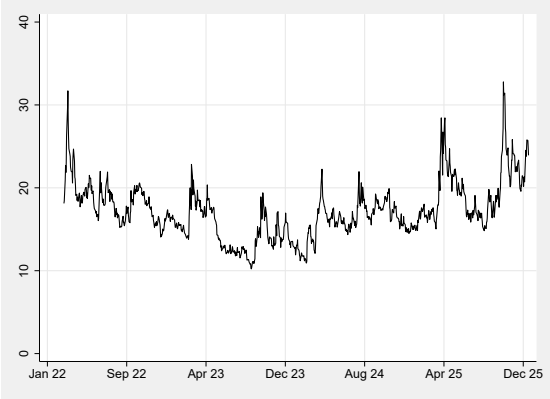
Note: The above Table presents the averaged connectedness measures between the analysed variables, estimated employing the Quantile Vector Autoregression (QVAR) methodology. Connectedness is estimated using the normalised Generalized Forecast Error Variance Decomposition (GFEVD) of Koop et al. [1996] and Pesaran and Shin [1998], as adjusted by Cunado et al. [2023]. TO denotes spillovers transmitted to the remaining variables, whereas FROM denotes spillovers received from them. NET stands for the net spillover impact, calculated as (TO-FROM). TCI is the Total Connectedness Index, which mirrors systemic risk; higher TCI values indicate stronger cross-variable spillovers, indicating a more tightly connected system with higher fragility.

Figure 1: Time-varying Performance of Selected Uncertainty Indices

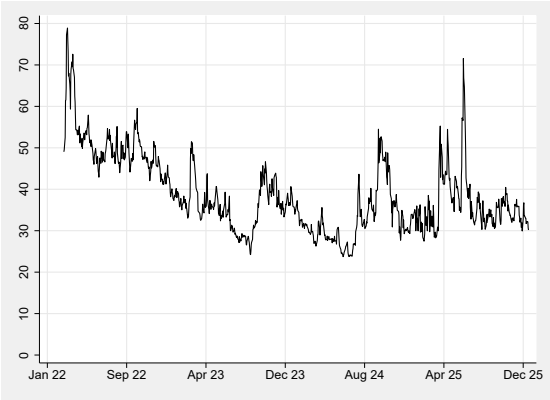
(a) VIX



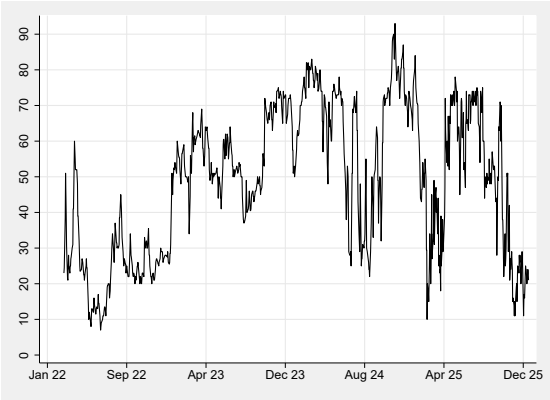
(b) Gold Volatility



(c) Oil Volatility



(d) Cryptocurrency Uncertainty Index



Note: This figure plots the daily values of the CBOE Volatility Index (VIX), the CBOE Gold ETF Volatility Index (GoldV), the CBOE Crude Oil ETF Volatility Index (OilV), and the Crypto Fear & Greed Index (CryptoS) over the period 24 February 2022 to 8 January 2026. VIX, GoldV, and OilV are obtained from the Federal Reserve Economic Data (FRED) database, while CryptoS is sourced from alternative.me.

Table 2: Estimated Averaged Dynamic Connectedness at the 5th Quantile (*continued*)

	VIX	IndexV	JPM	BoAm	Citi	W.Far	Gold.S	Morg.S	US Banc	Visa	FROM
						OilV and major banks					
OilV	-	18.89	10.19	10.21	9.92	11.10	9.58	9.79	9.84	10.37	61.11
JP Morgan	-	9.50	8.37	12.42	11.74	12.88	11.56	11.55	11.05	10.93	91.63
Bank of America	-	8.68	12.54	5.78	12.13	14.11	11.97	11.84	12.24	10.71	94.22
Citigroup	-	8.91	11.90	12.44	9.21	12.88	11.31	11.26	11.50	10.59	90.79
Wells Fargo	-	9.28	11.78	12.24	11.78	10.92	11.20	11.17	11.41	10.23	89.08
Goldman Sachs	-	8.53	12.01	12.29	11.83	13.11	8.45	12.01	11.22	10.54	91.55
Morgan Stanley	-	9.07	12.00	12.24	11.93	12.92	11.93	7.67	11.39	10.83	92.33
US Bancorp	-	9.45	11.54	12.18	11.64	13.13	11.26	11.30	8.73	10.78	91.27
Visa	-	10.13	11.47	11.51	11.27	12.02	11.12	11.06	10.98	10.44	89.56
TO	-	73.54	93.44	95.53	92.25	102.16	90.03	89.98	89.65	84.97	
NET	-	-7.56	1.80	1.30	1.47	13.07	-1.51	-2.34	-1.62	-4.60	TCI=90.17
						VIX, OilV and major banks					
VIX	15.01	11.18	9.05	9.00	8.98	9.19	9.64	9.35	9.32	9.28	84.99
OilV	10.82	9.98	9.74	9.41	9.72	9.83	10.27	10.00	9.81	10.41	90.02
JP Morgan	8.81	9.34	6.17	10.95	10.79	10.54	11.35	11.23	10.58	10.22	93.83
Bank of America	8.44	8.87	11.07	5.05	11.24	11.27	11.91	11.24	11.13	9.79	94.95
Citigroup	8.37	9.03	10.71	10.93	7.60	10.73	11.05	10.78	10.70	10.09	92.40
Wells Fargo	8.69	8.93	10.42	10.67	10.46	8.39	11.09	10.66	10.69	9.99	91.61
Goldman Sachs	8.65	8.79	10.34	10.77	10.38	10.55	8.93	11.12	10.43	10.03	91.07
Morgan Stanley	8.87	9.44	10.70	10.90	10.58	10.72	11.73	6.05	10.62	10.38	93.95
US Bancorp	9.08	9.16	10.57	10.77	10.63	11.27	10.95	10.57	6.68	10.30	93.32
Visa	9.40	9.77	10.23	10.29	10.10	10.32	11.16	10.70	10.38	7.66	92.34
TO	81.14	84.51	92.85	93.71	92.89	94.42	99.14	95.65	93.67	90.49	
NET	-3.85	-5.51	-0.98	-1.24	0.49	2.81	8.08	1.71	0.35	-1.86	TCI=91.85
						CryptoS and major banks					
CryptoS	-	16.81	11.12	9.71	10.26	11.27	10.15	10.08	10.27	10.36	83.19
JP Morgan	-	10.62	9.62	11.46	11.50	12.13	11.45	11.10	11.10	11.02	90.38
Bank of America	-	9.91	12.50	7.00	12.11	13.03	11.93	11.61	11.33	10.57	93.00
Citigroup	-	10.10	12.32	11.86	8.37	12.17	11.69	11.48	11.19	10.84	91.63
Wells Fargo	-	9.64	12.07	11.92	11.48	10.42	11.48	11.02	11.62	10.34	89.58
Goldman Sachs	-	10.49	12.38	11.67	11.69	12.21	8.12	11.69	10.98	10.77	91.88
Morgan Stanley	-	10.47	12.37	11.58	11.97	11.90	11.94	7.51	11.16	11.10	92.49
US Bancorp	-	10.14	11.97	11.11	11.48	12.22	11.05	10.95	10.77	10.31	89.23
Visa	-	10.36	11.70	10.83	11.13	11.68	11.29	11.14	10.87	10.98	89.02
TO	-	81.73	96.44	90.12	91.62	96.61	90.98	89.05	88.54	85.31	
NET	-	-1.46	6.05	-2.87	-0.01	7.03	-0.90	-3.44	-0.69	-3.71	TCI=90.04
						VIX, CryptoS and major banks					
VIX	19.52	9.68	9.55	8.57	8.62	8.93	8.57	8.78	8.79	9.00	80.48
CryptoS	9.94	11.80	10.54	9.51	9.79	9.74	9.32	10.05	9.57	9.75	88.20
JP Morgan	8.93	9.36	7.24	11.01	10.60	10.53	10.65	11.01	10.21	10.45	92.76
Bank of America	8.63	9.29	11.49	4.94	10.97	11.18	11.18	11.33	10.73	10.27	95.06
Citigroup	8.59	9.23	11.37	11.06	6.39	10.77	10.79	10.92	10.55	10.34	93.61
Wells Fargo	9.01	9.02	10.91	10.93	10.36	7.65	10.49	10.92	10.48	10.23	92.35
Goldman Sachs	8.71	9.11	11.14	10.84	10.47	10.79	7.39	11.25	10.24	10.07	92.61
Morgan Stanley	8.90	9.48	11.37	10.78	10.64	10.76	10.83	6.56	10.52	10.17	93.44
US Bancorp	9.51	9.07	10.93	10.61	10.54	10.73	10.36	10.92	7.03	10.31	92.97
Visa	9.39	9.61	11.01	10.33	10.28	10.44	10.35	10.70	10.36	7.53	92.47
TO	61.61	83.85	98.30	93.62	92.28	93.86	92.53	95.87	91.44	90.59	
NET	1.13	-4.35	5.54	-1.43	-1.33	1.50	-0.07	2.43	-1.53	-1.89	TCI=91.40

Note: The above Table presents the averaged connectedness measures between the analysed variables, estimated employing the Quantile Vector Autoregression (QVAR) methodology. Connectedness is estimated using the normalised Generalized Forecast Error Variance Decomposition (GFEVD) of Koop et al. [1996] and Pesaran and Shin [1998], as adjusted by Cunado et al. [2023]. TO denotes spillovers transmitted to the remaining variables, whereas FROM denotes spillovers received from them. NET stands for the net spillover impact, calculated as (TO-FROM). TCI is the Total Connectedness Index, which mirrors systemic risk; higher TCI values indicate stronger cross-variable spillovers, indicating a more tightly connected system with higher fragility.

Table 3: Estimated Averaged Dynamic Connectedness at the 50th Quantile

	VIX	IndexV	JPM	BoAm	Citi	W.Far	Gold.S	Morg.S	US Banc	Visa	FROM
						VIX and major banks					
VIX	46.66	-	6.95	6.37	6.94	5.01	7.68	7.67	6.24	6.48	53.34
JP Morgan	5.96	-	26.69	12.00	10.90	11.27	10.78	9.51	8.06	4.82	73.31
Bank of America	5.50	-	11.95	17.77	13.13	13.41	11.27	11.42	11.60	3.95	82.23
Citigroup	5.90	-	10.70	13.12	24.94	10.85	10.67	10.29	9.48	4.05	75.06
Wells Fargo	4.29	-	11.31	13.74	11.24	25.24	9.73	9.81	10.65	4.00	74.76
Goldman Sachs	6.74	-	10.95	11.20	10.84	9.84	24.34	13.08	8.63	4.38	75.66
Morgan Stanley	6.51	-	9.41	11.32	10.53	9.76	12.98	25.48	9.32	4.69	74.52
US Bancorp	5.73	-	8.20	11.92	9.65	10.73	8.84	9.47	31.30	4.16	68.70
Visa	7.21	-	6.25	4.59	5.17	4.68	5.45	5.69	4.91	56.06	43.94
TO	47.84	-	75.72	84.26	78.41	75.56	77.40	76.92	68.88	36.53	
NET	-5.51	-	2.41	2.03	3.35	0.80	1.74	2.40	0.18	-7.40	TCI=69.06
						GoldV and major banks					
GoldV	-	72.17	3.05	3.74	3.69	2.96	3.64	3.95	3.69	3.12	27.83
JP Morgan	-	2.07	27.29	12.55	11.40	11.79	11.13	10.09	8.48	5.19	72.71
Bank of America	-	2.48	12.36	17.47	13.67	13.99	11.79	12.01	12.13	4.11	82.53
Citigroup	-	2.43	11.12	13.67	25.59	11.16	11.06	10.83	9.97	4.16	74.41
Wells Fargo	-	1.89	11.59	14.15	11.42	25.64	10.15	10.22	10.88	4.07	74.36
Goldman Sachs	-	2.50	11.45	11.91	11.39	10.50	24.53	13.94	9.14	4.64	75.47
Morgan Stanley	-	2.38	9.93	11.97	11.12	10.33	13.76	25.83	9.74	4.94	74.17
US Bancorp	-	2.80	8.60	12.51	10.21	11.17	9.29	9.89	31.31	4.20	68.69
Visa	-	2.34	6.67	4.80	5.34	4.76	5.90	6.04	5.07	59.08	40.92
TO	-	18.89	74.78	85.29	78.25	76.67	76.72	76.96	69.11	34.42	
NET	-	-8.94	2.07	2.75	3.84	2.31	1.25	2.79	0.42	-6.49	TCI=65.68
						VIX, GoldV and major banks					
VIX	40.95	5.71	6.93	6.38	6.90	5.03	7.65	7.74	6.21	6.51	59.05
GoldV	8.69	63.47	3.13	3.74	3.76	3.02	3.57	3.97	3.65	3.01	36.53
JP Morgan	5.77	2.00	26.21	11.72	10.67	11.03	10.40	9.38	7.98	4.84	73.79
Bank of America	5.32	2.32	11.61	17.58	12.76	13.10	10.99	11.23	11.29	3.81	82.42
Citigroup	5.67	2.23	10.46	12.72	24.92	10.45	10.38	10.07	9.22	3.86	75.08
Wells Fargo	4.19	1.81	11.13	13.55	10.98	24.61	9.67	9.75	10.44	3.87	75.39
Goldman Sachs	6.53	2.30	10.63	11.00	10.62	9.72	23.73	12.85	8.40	4.23	76.27
Morgan Stanley	6.37	2.20	9.16	11.03	10.24	9.54	12.68	25.28	8.99	4.52	74.72
US Bancorp	5.58	2.58	8.10	11.59	9.45	10.48	8.57	9.22	10.46	3.97	69.54
Visa	7.16	2.24	6.28	4.52	5.07	4.53	5.54	5.74	4.80	54.13	45.87
TO	55.28	23.38	77.44	86.25	80.44	76.90	79.45	79.94	70.98	38.62	
NET	-3.77	-13.16	3.64	3.83	5.36	1.51	3.18	5.22	1.44	-7.26	TCI=66.87

(continued on the next page)

Note: The above Table presents the averaged connectedness measures between the analysed variables, estimated employing the Quantile Vector Autoregression (QVAR) methodology. Connectedness is estimated using the normalised Generalized Forecast Error Variance Decomposition (GFEVD) of Koop et al. [1996] and Pesaran and Shin [1998], as adjusted by Cunado et al. [2023]. TO denotes spillovers transmitted to the remaining variables, whereas FROM denotes spillovers received from them. NET stands for the net spillover impact, calculated as (TO-FROM). TCI is the Total Connectedness Index, which mirrors systemic risk; higher TCI values indicate stronger cross-variable spillovers, indicating a more tightly connected system with higher fragility.

Table 3: Estimated Averaged Dynamic Connectedness at the 50th Quantile (*continued*)

	VIX	IndexV	JPM	BoAm	Citi	W.Far	Gold.S	Morg.S	US Banc	Visa	FROM
						OilV and major banks					
OilV	-	70.20	2.82	3.81	3.86	3.18	5.26	4.57	3.80	2.50	29.80
JP Morgan	-	1.43	27.51	12.65	11.52	11.94	11.30	10.04	8.43	5.18	72.49
Bank of America	-	1.69	12.54	17.24	13.94	14.21	11.96	11.99	12.17	4.26	82.76
Citigroup	-	1.71	11.34	14.08	24.70	111.52	11.26	10.96	10.14	4.30	75.30
Wells Fargo	-	1.34	11.74	14.31	11.65	25.28	10.14	10.28	10.95	4.31	74.72
Goldman Sachs	-	2.09	11.61	12.03	11.45	10.50	24.44	13.96	9.15	4.77	75.56
Morgan Stanley	-	1.84	10.01	11.95	11.18	10.35	13.83	25.97	9.86	5.02	74.03
US Bancorp	-	1.73	8.55	12.57	10.24	11.24	9.30	9.95	32.08	4.35	67.92
Visa	-	1.16	6.66	4.88	5.41	4.89	5.75	5.99	5.14	60.12	39.88
TO	-	13.01	75.27	86.28	79.26	77.82	78.79	77.71	69.64	34.68	
NET	-	-16.80	2.78	3.52	3.96	3.11	3.23	3.68	1.72	-5.20	TCI=65.83
						VIX, OilV and major banks					
VIX	43.46	3.75	6.93	6.29	6.87	4.93	7.61	7.56	6.16	6.44	56.54
OilV	6.52	64.79	2.65	3.67	3.72	3.00	5.10	4.38	3.71	2.40	35.21
JP Morgan	5.90	1.34	26.28	11.85	10.81	11.16	10.61	9.33	77.93	4.76	73.72
Bank of America	5.36	1.63	11.74	17.24	13.05	13.32	11.12	11.21	11.39	33.93	82.76
Citigroup	5.82	1.65	10.58	13.10	23.98	10.75	10.54	10.22	9.38	3.99	76.02
Wells Fargo	4.24	1.29	11.21	13.71	11.17	24.41	9.63	9.77	10.55	4.01	75.59
Goldman Sachs	6.59	1.94	10.76	11.04	10.62	9.60	23.86	12.81	8.41	4.31	76.14
Morgan Stanley	6.36	1.78	9.19	11.09	10.38	9.63	12.75	25.01	9.16	4.69	74.99
US Bancorp	5.63	1.68	8.09	11.78	9.56	10.62	8.69	9.31	30.48	4.15	69.52
Visa	7.23	1.28	6.21	4.55	5.14	4.68	5.40	5.50	4.92	54.84	45.16
TO	53.65	16.35	77.35	87.08	81.34	77.77	81.43	80.34	71.60	38.74	
NET	-2.89	-18.86	3.63	4.32	5.31	2.18	5.29	5.35	2.09	-6.42	TCI=66.56
						CryptoS and major banks					
CryptoS	-	79.30	1.96	2.23	2.85	2.26	2.86	4.16	2.02	2.34	20.70
JP Morgan	-	1.00	27.32	12.82	11.60	12.15	11.43	10.18	8.41	5.14	72.68
Bank of America	-	0.95	12.72	17.65	13.92	14.29	11.93	12.05	12.19	4.31	82.35
Citigroup	-	0.94	11.38	13.89	25.52	11.49	11.39	10.94	10.09	4.95	74.48
Wells Fargo	-	0.80	11.93	14..36	11.71	25.26	10.28	10.35	11.01	4.30	74.74
Goldman Sachs	-	0.97	11.79	12.10	111.73	10.74	24.23	14.21	9.36	4.89	75.77
Morgan Stanley	-	1.19	10.23	12.19	11.36	10.64	14.05	25.17	10.03	5.14	74.83
US Bancorp	-	0.85	8.60	12.62	10.35	11.32	0.46	10.03	32.39	4.37	67.61
Visa	-	1.30	6.65	4.95	5.56	5.00	5.84	5.99	5.12	59.62	40.38
TO	-	7.95	75.27	85.16	79.07	77.68	77.24	77.93	68.22	34.83	
NET	-	-12.75	2.59	2.81	4.59	3.14	1.47	3.09	0.61	-5.56	TCI=64.84
						VIX, CryptoS and major banks					
VIX	45.54	0.95	7.03	6.41	7.09	5.00	7.73	7.59	6.22	6.48	54.45
CryptoS	3.21	76.27	2.04	2.15	2.77	2.17	2.93	4.04	2.06	2.36	23.73
JP Morgan	5.93	0.80	25.86	12.06	10.87	11.36	10.72	9.54	7.91	4.78	74.14
Bank of America	5.49	0.88	11.97	17.57	12.99	13.30	11.11	11.26	11.41	4.02	82.43
Citigroup	5.95	0.90	10.51	12.91	24.79	10.66	10.59	10.19	9.33	4.08	75.21
Wells Fargo	4.29	0.79	11.40	13.69	11.15	24.57	9.74	9.82	10.52	4.03	75.43
Goldman Sachs	6.78	0.91	10.89	11.13	10.81	9.87	23.54	13.09	8.52	4.45	76.46
Morgan Stanley	6.40	1.07	9.50	11.24	10.51	9.81	12.98	24.55	9.21	4.74	75.45
US Bancorp	5.74	0.84	8.08	11.79	9.59	10.59	8.73	9.31	31.18	4.15	68.82
Visa	7.25	1.27	6.34	4.69	5.33	4.79	5.53	5.72	4.90	54.17	45.83
TO	51.04	8.60	77.84	66.08	81.07	77.57	80.06	80.56	70.08	39.08	
NET	-3.42	-15.10	3.70	3.65	5.86	2.13	3.60	5.11	1.26	-6.75	TCI=65.20

Note: The above Table presents the averaged connectedness measures between the analysed variables, estimated employing the Quantile Vector Autoregression (QVAR) methodology. Connectedness is estimated using the normalised Generalized Forecast Error Variance Decomposition (GFEVD) of Koop et al. [1996] and Pesaran and Shin [1998], as adjusted by Cunado et al. [2023]. TO denotes spillovers transmitted to the remaining variables, whereas FROM denotes spillovers received from them. NET stands for the net spillover impact, calculated as (TO-FROM). TCI is the Total Connectedness Index, which mirrors systemic risk; higher TCI values indicate stronger cross-variable spillovers, indicating a more tightly connected system with higher fragility.

Table 4: Estimated Averaged Dynamic Connectedness at the 95th Quantile

	VIX	IndexV	JPM	BoAm	Citi	W.Far	Gold.S	Morg.S	US Banc	Visa	FROM
						VIX and major banks					
VIX	19.18	-	9.88	9.94	10.68	10.43	10.75	9.93	9.43	9.78	80.82
JP Morgan	11.17	-	6.57	11.96	12.28	12.52	12.61	11.50	10.80	10.58	93.43
Bank of America	11.47	-	11.10	9.03	11.85	11.87	12.21	11.20	10.83	10.44	90.97
Citigroup	11.43	-	11.69	12.15	5.97	12.38	12.77	11.57	11.31	10.73	94.03
Wells Fargo	11.26	-	11.87	12.08	12.33	6.48	12.69	11.49	11.13	10.67	93.52
Goldman Sachs	10.93	-	11.42	11.53	11.92	11.81	9.81	11.41	10.79	10.58	90.39
Morgan Stanley	10.96	-	11.73	11.59	12.38	12.25	12.92	6.18	11.17	10.82	93.82
US Bancorp	11.38	-	11.17	12.01	11.95	12.30	12.58	11.45	6.70	10.44	93.30
Visa	11.43	-	11.08	11.05	11.55	11.56	12.12	10.96	10.41	9.84	90.16
TO	90.04	-	89.94	92.30	94.95	95.12	98.64	89.53	85.86	84.04	
NET	9.22	-	-3.49	1.34	0.92	1.60	8.26	-4.30	-7.44	-6.11	TCI=91.16
						GoldV and major banks					
GoldV	-	17.16	9.60	9.81	10.11	10.71	11.60	10.22	10.40	10.40	82.84
JP Morgan	-	9.57	8.00	11.26	11.82	12.54	13.11	11.62	11.02	11.05	92.00
Bank of America	-	9.59	11.24	9.25	11.91	12.01	12.57	11.57	11.27	10.60	90.75
Citigroup	-	10.16	11.59	11.64	6.42	12.45	13.45	11.70	11.66	10.93	93.58
Wells Fargo	-	9.85	11.87	11.64	11.94	7.65	13.03	11.58	11.69	10.75	92.35
Goldman Sachs	-	9.96	11.48	11.33	12.00	12.10	9.14	11.87	11.31	10.82	90.86
Morgan Stanley	-	9.91	11.41	11.52	11.85	12.09	13.53	7.29	11.48	10.93	92.71
US Bancorp	-	9.90	11.00	11.49	11.75	12.20	13.05	11.71	8.11	10.70	91.89
Visa	-	10.77	10.47	10.36	11.00	11.37	11.75	10.90	10.40	12.98	87.02
TO	-	79.71	88.73	89.06	92.38	95.45	102.09	91.16	89.22	86.18	
NET	-	-3.13	-3.27	-1.70	-1.20	3.11	11.23	-1.55	-2.67	-0.84	TCI=90.44
						VIX, GoldV and major banks					
VIX	10.24	10.33	9.65	9.30	10.56	10.08	10.98	9.83	9.35	9.68	89.76
GoldV	11.20	7.27	9.88	9.72	10.56	10.47	11.21	10.14	9.70	9.85	92.73
JP Morgan	9.43	8.57	6.16	10.50	11.51	11.27	11.66	10.76	10.26	9.89	93.84
Bank of America	9.80	8.90	10.35	6.79	11.43	10.67	11.29	10.73	10.24	9.81	93.21
Citigroup	9.64	8.80	10.98	10.65	5.12	11.17	12.04	10.87	10.42	10.32	94.88
Wells Fargo	9.67	8.91	11.01	10.50	11.41	5.61	11.83	10.71	10.47	9.89	94.39
Goldman Sachs	9.49	8.57	10.67	10.21	11.21	10.75	8.51	10.68	10.01	9.89	91.49
Morgan Stanley	9.48	8.84	10.65	10.54	11.44	10.86	11.90	5.76	10.35	10.17	94.24
US Bancorp	9.73	8.89	10.40	10.46	11.26	11.12	11.39	10.66	6.43	9.64	93.57
Visa	9.95	9.26	10.11	9.69	10.61	10.63	11.22	10.31	9.78	8.43	91.57
TO	88.39	81.07	93.69	91.58	100.00	97.02	103.51	94.69	90.60	89.12	
NET	-1.37	-11.66	-0.15	-1.82	5.13	2.63	12.02	0.44	-2.97	-2.45	TCI=92.97

(continued on the next page)

Note: The above Table presents the averaged connectedness measures between the analysed variables, estimated employing the Quantile Vector Autoregression (QVAR) methodology. Connectedness is estimated using the normalised Generalized Forecast Error Variance Decomposition (GFEVD) of Koop et al. [1996] and Pesaran and Shin [1998], as adjusted by Cunado et al. [2023]. TO denotes spillovers transmitted to the remaining variables, whereas FROM denotes spillovers received from them. NET stands for the net spillover impact, calculated as (TO-FROM). TCI is the Total Connectedness Index, which mirrors systemic risk; higher TCI values indicate stronger cross-variable spillovers, indicating a more tightly connected system with higher fragility.

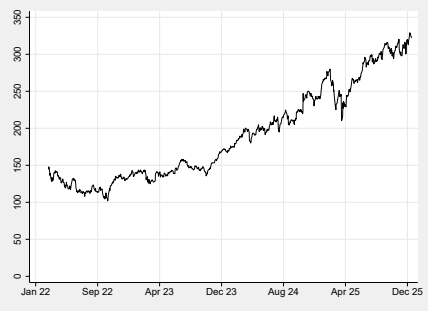
Table 4: Estimated Averaged Dynamic Connectedness at the 95th Quantile (*continued*)

	VIX	IndexV	JPM	BoAm	Citi	W.Far	Gold.S	Morg.S	US Banc	Visa	FROM
						OilV and major banks					
OilV	-	18.14	9.77	10.88	10.31	10.56	10.53	9.90	9.41	10.52	81.86
JP Morgan	-	8.92	9.11	12.49	12.12	12.56	12.18	11.51	10.32	10.80	90.89
Bank of America	-	9.08	11.59	10.78	11.84	11.97	12.06	11.39	10.57	10.73	89.22
Citigroup	-	9.09	11.74	12.81	8.83	12.23	12.18	11.53	10.74	10.85	91.17
Wells Fargo	-	9.20	11.92	12.83	12.12	8.90	12.09	11.55	10.87	10.53	91.10
Goldman Sachs	-	9.13	11.49	12.61	12.08	11.94	9.13	11.93	10.75	10.94	90.87
Morgan Stanley	-	9.22	11.48	12.68	12.06	12.25	13.05	0.39	10.84	11.02	92.61
US Bancorp	-	9.36	11.21	12.27	12.02	12.14	12.10	11.70	8.41	10.79	91.59
Visa	-	9.99	10.97	11.87	11.20	11.27	11.46	11.19	10.19	12.06	87.94
TO	-	74.00	90.17	98.24	93.74	94.90	95.65	90.69	83.69	86.15	
NET	-	-7.87	-0.72	9.02	2.57	3.80	4.79	-1.92	-7.90	-1.78	TCI=89.69
						VIX, OilV and major banks					
VIX	11.70	10.77	9.23	10.11	9.97	10.21	10.01	9.54	9.17	9.29	88.30
OilV	11.94	7.91	9.57	10.80	10.14	10.46	10.24	9.70	9.62	9.62	92.09
JP Morgan	9.94	9.26	5.42	11.45	11.31	11.43	11.15	10.61	9.87	9.57	94.58
Bank of America	10.24	9.31	10.19	8.56	10.75	10.60	10.85	10.37	9.75	9.37	91.44
Citigroup	10.14	9.13	10.72	11.36	5.85	11.28	11.28	10.59	10.09	9.57	94.15
Wells Fargo	10.02	9.17	10.46	11.47	10.99	7.29	10.93	10.29	9.97	9.42	92.71
Goldman Sachs	10.21	9.21	10.45	11.05	10.85	10.93	6.94	10.73	9.94	9.69	93.06
Morgan Stanley	10.04	9.47	10.59	11.30	11.08	10.95	11.48	5.11	10.13	9.84	94.89
US Bancorp	10.28	9.49	10.10	11.12	10.86	11.07	10.99	10.50	5.91	9.68	94.09
Visa	10.53	9.61	9.95	10.90	10.55	10.57	10.59	10.05	9.61	7.66	92.34
TO	93.33	85.40	91.27	99.55	96.50	97.50	97.51	92.38	88.15	86.05	
NET	5.03	-6.69	-3.31	8.11	2.35	4.79	4.45	-2.51	-5.94	-6.29	TCI=92.76
						CryptoS and major banks					
CryptoS	-	14.36	10.70	10.87	11.25	10.45	11.33	10.30	10.26	10.48	85.64
JP Morgan	-	8.86	10.15	11.81	12.06	12.34	12.16	11.26	10.55	11.01	89.85
Bank of America	-	8.96	11.64	11.42	12.36	11.90	11.73	10.79	10.98	10.14	88.58
Citigroup	-	9.15	12.37	12.62	8.82	12.03	12.41	11.10	11.05	10.44	91.18
Wells Fargo	-	8.80	12.18	12.43	12.27	9.54	12.06	11.25	11.29	10.17	90.46
Goldman Sachs	-	8.62	12.36	12.01	12.31	11.68	9.64	11.88	10.89	10.60	90.36
Morgan Stanley	-	8.95	12.51	11.88	12.19	12.00	12.99	7.33	11.17	10.98	92.67
US Bancorp	-	8.87	11.79	12.24	11.97	11.90	11.92	11.06	9.63	10.63	90.37
Visa	-	9.07	11.89	10.83	11.09	10.84	11.53	10.58	10.25	13.95	86.05
TO	-	71.08	95.44	94.68	95.50	93.23	96.13	88.19	86.45	84.45	
NET	-	-14.56	5.59	6.11	4.32	2.77	5.76	-4.48	-3.92	-1.60	TCI=89.46
						VIX, CryptoS and major banks					
VIX	16.61	8.72	8.92	8.79	9.82	9.42	9.92	9.50	8.85	9.45	83.39
CryptoS	11.52	7.94	9.45	9.87	10.63	10.03	10.49	10.25	9.78	10.04	92.06
JP Morgan	10.30	8.81	6.45	10.76	11.12	10.99	10.78	10.55	10.14	10.10	93.55
Bank of America	10.56	8.94	10.05	7.18	11.08	10.67	10.83	10.51	10.27	9.91	92.82
Citigroup	10.46	8.91	10.30	10.83	6.40	10.88	11.18	10.67	10.33	10.04	93.60
Wells Fargo	10.51	8.95	10.38	10.74	11.27	6.63	10.65	10.53	10.37	9.95	93.37
Goldman Sachs	10.50	8.82	10.29	10.30	11.18	10.67	7.23	10.85	10.11	10.05	92.77
Morgan Stanley	10.31	9.14	10.55	10.55	11.22	10.74	11.23	5.69	10.28	10.30	94.31
US Bancorp	10.58	9.15	9.93	10.85	11.05	10.97	10.88	10.56	6.14	9.89	93.86
Visa	11.09	9.06	10.11	9.73	10.51	10.29	10.53	10.18	9.76	8.74	91.26
TO	95.82	80.51	89.99	92.42	97.88	94.66	96.50	93.61	89.88	89.73	
NET	12.43	-11.55	-3.57	-0.40	4.28	1.29	3.73	-0.70	-3.98	-1.53	TCI=92.10

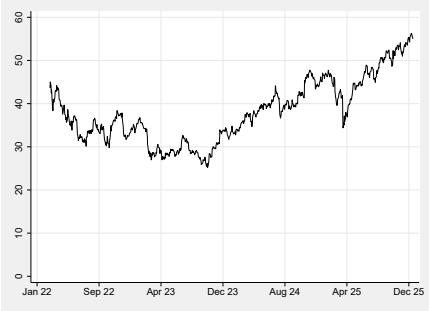
Note: The above Table presents the averaged connectedness measures between the analysed variables, estimated employing the Quantile Vector Autoregression (QVAR) methodology. Connectedness is estimated using the normalised Generalized Forecast Error Variance Decomposition (GFEVD) of Koop et al. [1996] and Pesaran and Shin [1998], as adjusted by Cunado et al. [2023]. TO denotes spillovers transmitted to the remaining variables, whereas FROM denotes spillovers received from them. NET stands for the net spillover impact, calculated as (TO-FROM). TCI is the Total Connectedness Index, which mirrors systemic risk; higher TCI values indicate stronger cross-variable spillovers, indicating a more tightly connected system with higher fragility.

Figure 2: Share Prices of Selected Analysed Financial Institutions

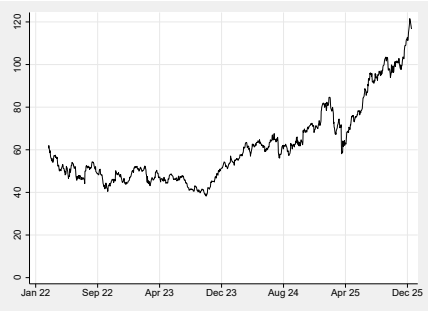
(a) JP Morgan



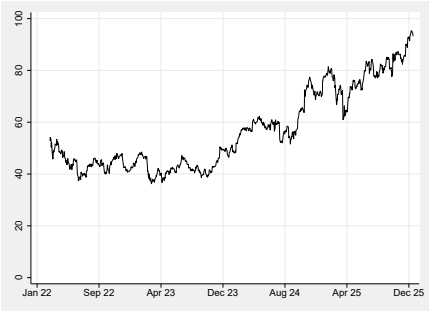
(b) Bank of America



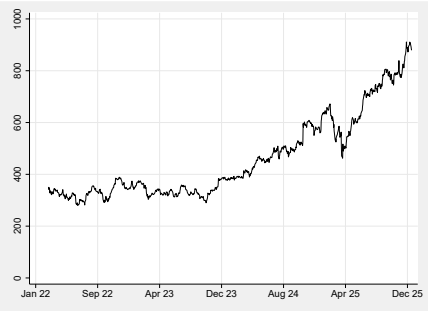
(c) Citigroup



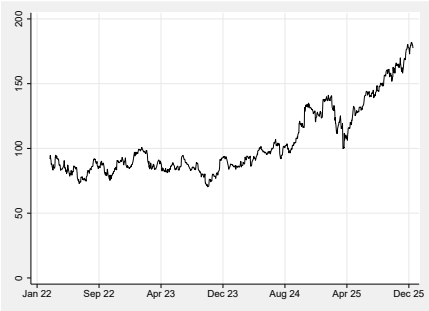
(d) Wells Fargo



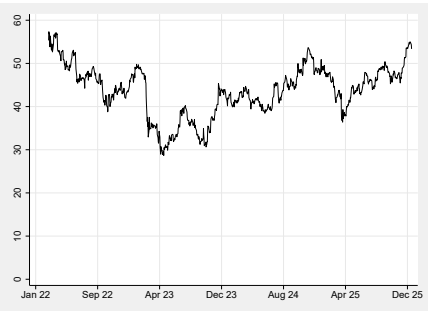
(e) Goldman Sachs



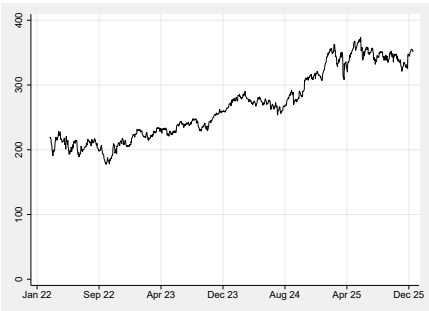
(f) Morgan Stanley



(g) US Bancorp



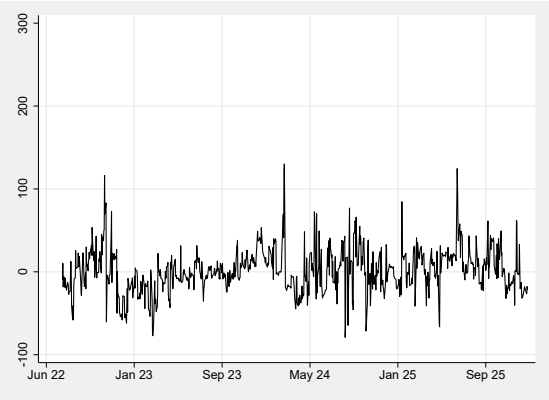
(h) Visa



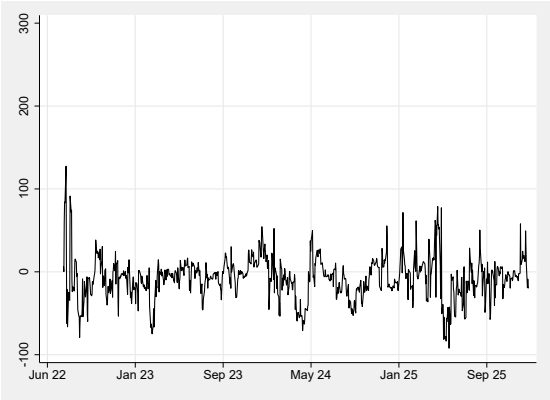
Note: This figure plots the daily share prices of JP Morgan Chase, Bank of America, Citigroup, Wells Fargo, Goldman Sachs, Morgan Stanley, US Bancorp, and Visa over the period 24 February 2022 to 8 January 2026. All series are retrieved from Datastream.

Figure 3: Net Extended Joint Connectedness with Financial Institutions at the 5th Percentile

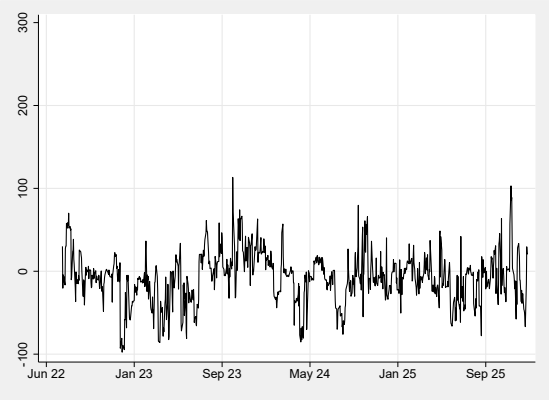
(a) VIX & Banks



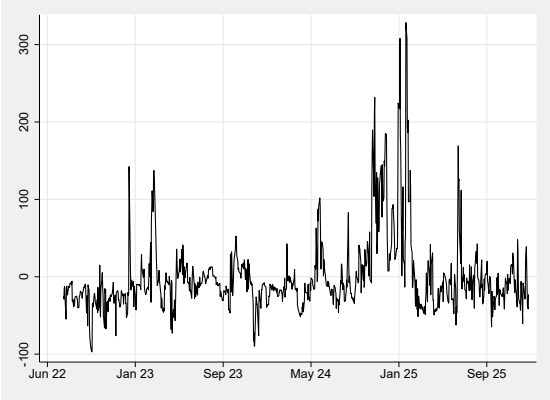
(b) GoldV & Banks



(c) OilV & Banks



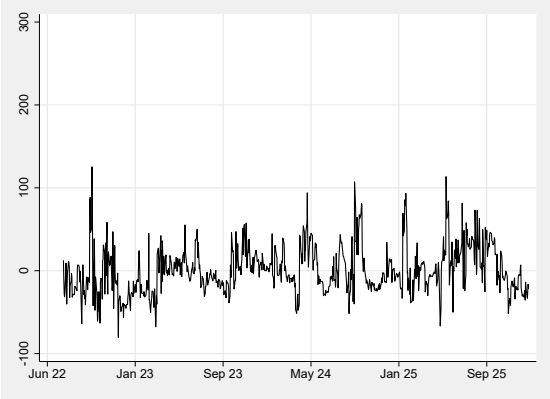
(d) CryptoS & Banks



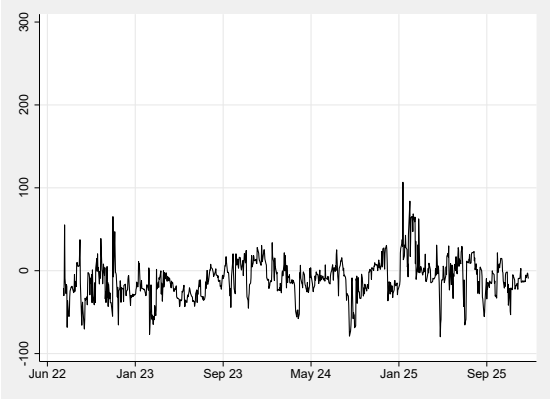
Note: The Net Extended Joint Connectedness (NEJC) measure summarizes the overall systemic role of each variable within the connectedness network by jointly accounting for its transmission and reception of shocks across all horizons considered. Formally, the total directional connectedness *to* others captures the cumulative impact of shocks originating in variable i on all remaining variables j, and is defined as: $C^g_{i\rightarrow\bullet}(H) = \sum_{\substack{j=1\\ j\neq i}}^{k} \psi\widetilde{\psi}^g_{ji}(H)$, where $\psi\widetilde{\psi}^g_{ji}(H)$ denotes the generalized forecast error variance decomposition at horizon H measuring the contribution of shocks in i to the forecast error variance of j. Conversely, the total directional connectedness *from* others measures the aggregate influence of shocks from all remaining variables j on variable i, and is given by $C^g_{i\leftarrow\bullet}(H) = \sum_{\substack{j=1\\ j\neq i}}^{k} \psi\widetilde{\psi}^g_{ij}(H)$. The Net Extended Joint Connectedness for variable i is then defined as the difference between these two directional components, $\text{NEJC}^g_i(H) = C^g_{i\rightarrow\bullet}(H) - C^g_{i\leftarrow\bullet}(H)$, and reflects the extent to which variable i acts as a net transmitter or a net receiver of shocks within the system. Positive values of $\text{NEJC}^g_i(H)$ indicate that variable i contributes more strongly to the propagation of shocks across the network than it absorbs, while negative values imply that it is predominantly influenced by shocks originating elsewhere. By aggregating spillovers across all variables and horizons, the NEJC measure provides a comprehensive characterization of each variable's extended joint contribution to systemic connectedness.

Figure 3: Net Extended Joint Connectedness with Financial Institutions at the 5th Percentile (*continued*)

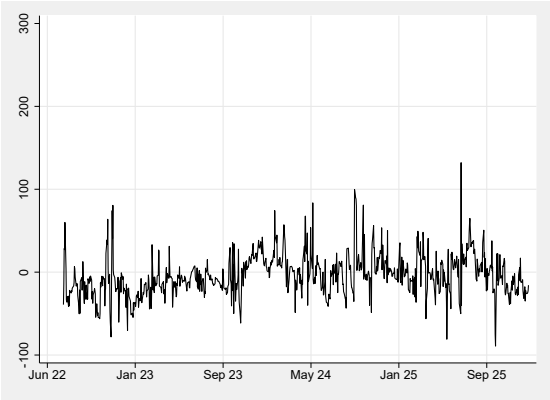
(e) VIX & (GoldV + Banks)



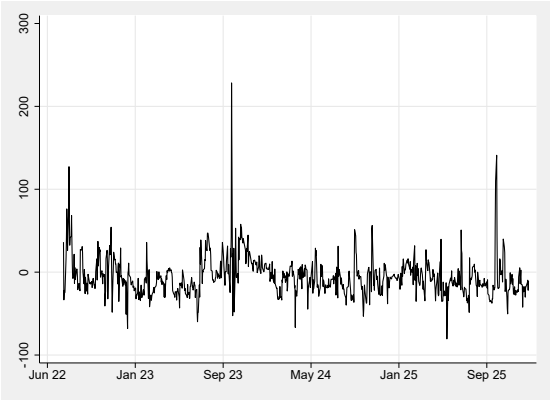
(f) GoldV & (VIX + Banks)



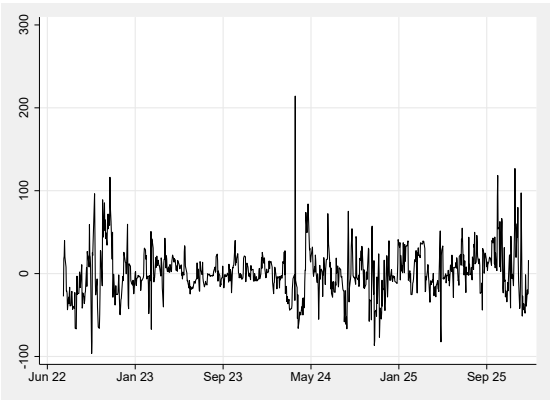
(g) VIX & (OilV + Banks)



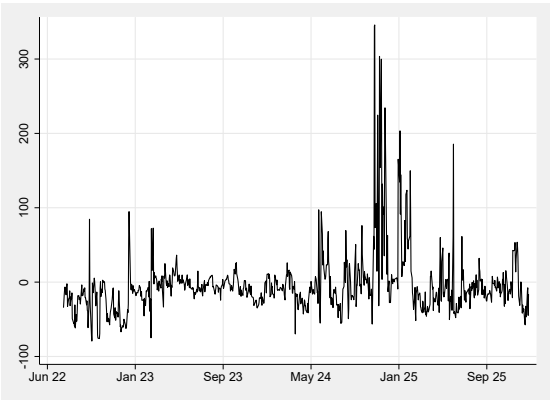
(h) OilV & (VIX + Banks)



(i) VIX & (CryptoS + Banks)



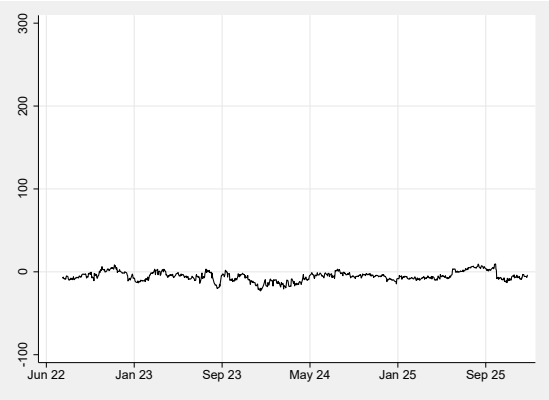
(j) CryptoS & (VIX + Banks)



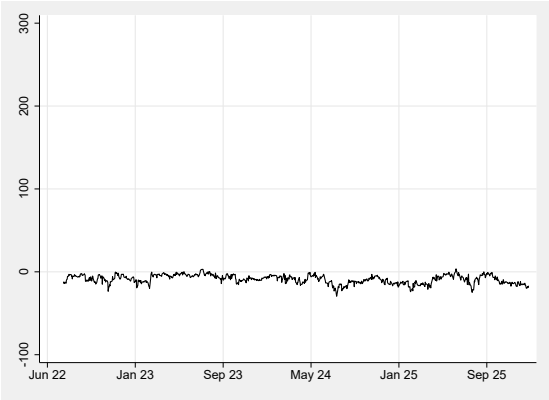
Note: The Net Extended Joint Connectedness (NEJC) measure summarizes the overall systemic role of each variable within the connectedness network by jointly accounting for its transmission and reception of shocks across all horizons considered. Formally, the total directional connectedness *to* others captures the cumulative impact of shocks originating in variable i on all remaining variables j, and is defined as: $C^g_{i\to\bullet}(H) = \sum_{\substack{j=1\\ j\neq i}}^{35} \psi\tilde{\psi}^g_{ji}(H)$, where $\psi\tilde{\psi}^g_{ji}(H)$ denotes the generalized forecast error variance decomposition at horizon H measuring the contribution of shocks in i to the forecast error variance of j. Conversely, the total directional connectedness *from* others measures the aggregate influence of shocks from all remaining variables j on variable i, and is given by $C^g_{i\leftarrow\bullet}(H) = \sum_{\substack{j=1\\ j\neq i}}^{k} \psi\tilde{\psi}^g_{ij}(H)$. The Net Extended Joint Connectedness for variable i is then defined as the difference between these two directional components, $\text{NEJC}^g_i(H) = C^g_{i\to\bullet}(H) - C^g_{i\leftarrow\bullet}(H)$, and reflects the extent to which variable i acts as a net transmitter or a net receiver of shocks within the system. Positive values of $\text{NEJC}^g_i(H)$ indicate that variable i contributes more strongly to the propagation of shocks across the network than it absorbs, while negative values imply that it is predominantly influenced by shocks originating elsewhere. By aggregating spillovers across all variables and horizons, the NEJC measure provides a comprehensive characterization of each variable's extended joint contribution to systemic connectedness.

Figure 4: Net Extended Joint Connectedness with Financial Institutions at the 50th Percentile

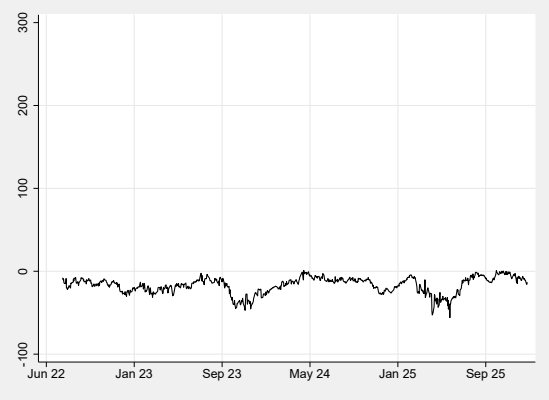
(a) VIX & Banks



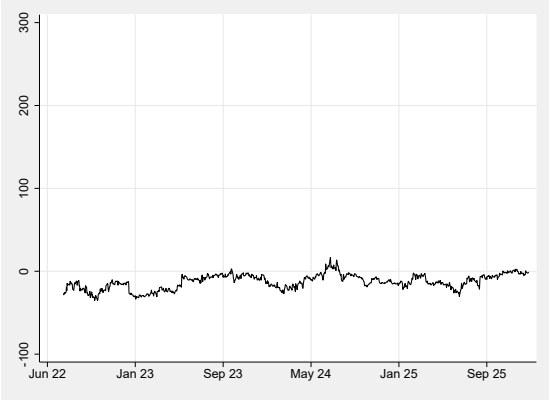
(b) GoldV & Banks



(c) OilV & Banks



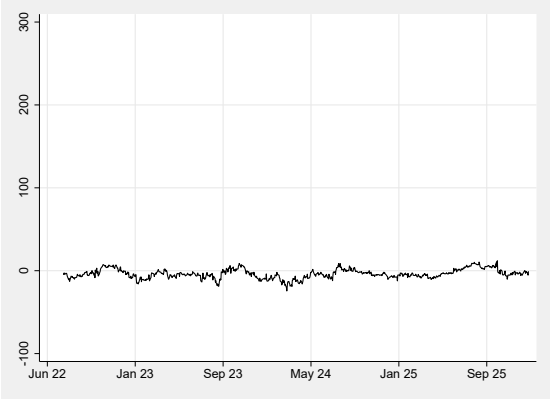
(d) CryptoS & Banks



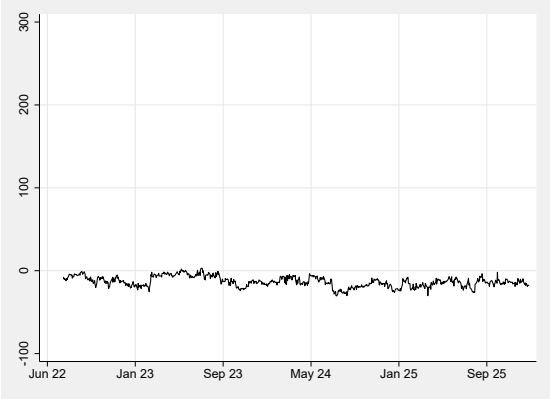
Note: The Net Extended Joint Connectedness (NEJC) measure summarizes the overall systemic role of each variable within the connectedness network by jointly accounting for its transmission and reception of shocks across all horizons considered. Formally, the total directional connectedness *to* others captures the cumulative impact of shocks originating in variable i on all remaining variables j, and is defined as: $C^g_{i\rightarrow\bullet}(H) = \sum_{\substack{j=1\\ j\neq i}}^{k} \psi\widetilde{\psi}^g_{ji}(H)$, where $\psi\widetilde{\psi}^g_{ji}(H)$ denotes the generalized forecast error variance decomposition at horizon H measuring the contribution of shocks in i to the forecast error variance of j. Conversely, the total directional connectedness *from* others measures the aggregate influence of shocks from all remaining variables j on variable i, and is given by $C^g_{i\leftarrow\bullet}(H) = \sum_{\substack{j=1\\ j\neq i}}^{k} \psi\widetilde{\psi}^g_{ij}(H)$. The Net Extended Joint Connectedness for variable i is then defined as the difference between these two directional components, $\text{NEJC}^g_i(H) = C^g_{i\rightarrow\bullet}(H) - C^g_{i\leftarrow\bullet}(H)$, and reflects the extent to which variable i acts as a net transmitter or a net receiver of shocks within the system. Positive values of $\text{NEJC}^g_i(H)$ indicate that variable i contributes more strongly to the propagation of shocks across the network than it absorbs, while negative values imply that it is predominantly influenced by shocks originating elsewhere. By aggregating spillovers across all variables and horizons, the NEJC measure provides a comprehensive characterization of each variable's extended joint contribution to systemic connectedness.

Figure 4: Net Extended Joint Connectedness with Financial Institutions at the 50th Percentile (*continued*)

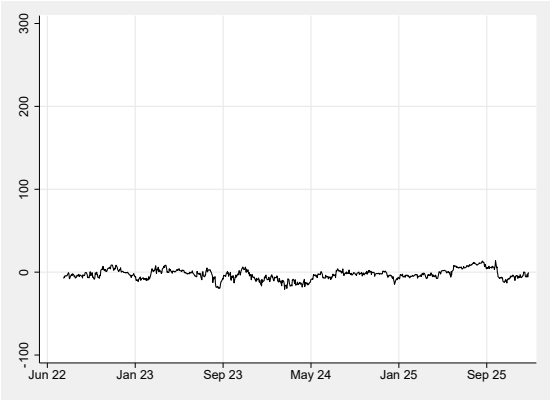
(e) VIX & (GoldV + Banks)



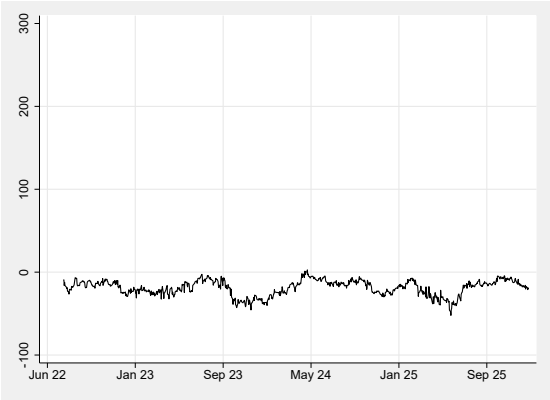
(f) GoldV & (VIX + Banks)



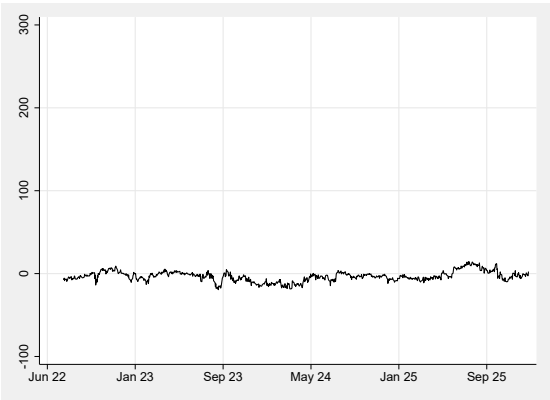
(g) VIX & (OilV + Banks)



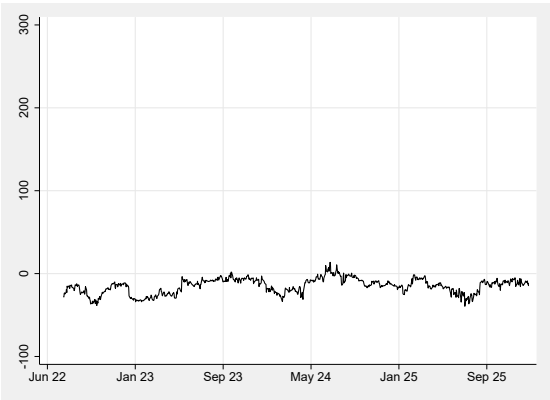
(h) OilV & (VIX + Banks)



(i) VIX & (CryptoS + Banks)



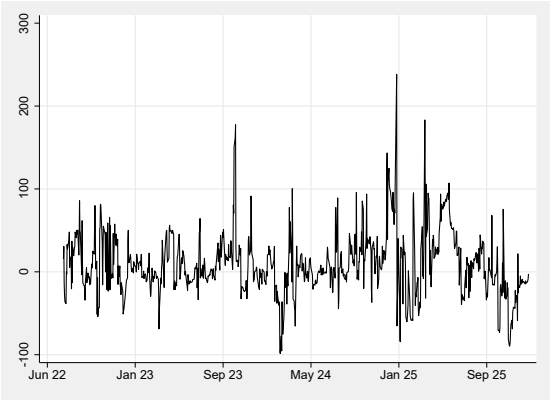
(j) CryptoS & (VIX + Banks)



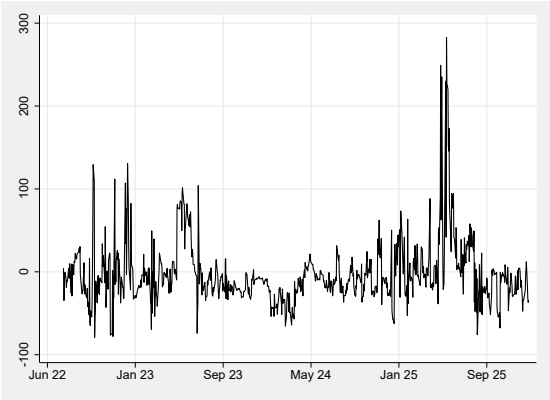
Note: The Net Extended Joint Connectedness (NEJC) measure summarizes the overall systemic role of each variable within the connectedness network by jointly accounting for its transmission and reception of shocks across all horizons considered. Formally, the total directional connectedness *to* others captures the cumulative impact of shocks originating in variable i on all remaining variables j, and is defined as: $C^g_{i\rightarrow\bullet}(H) = \sum_{\substack{j=1\\ j\neq i}}^{k} \psi\tilde{\psi}^g_{ji}(H)$, where $\psi\tilde{\psi}^g_{ji}(H)$ denotes the generalized forecast error variance decomposition at horizon H measuring the contribution of shocks in i to the forecast error variance of j. Conversely, the total directional connectedness *from* others measures the aggregate influence of shocks from all remaining variables j on variable i, and is given by $C^g_{i\leftarrow\bullet}(H) = \sum_{\substack{j=1\\ j\neq i}}^{k} \psi\tilde{\psi}^g_{ij}(H)$. The Net Extended Joint Connectedness for variable i is then defined as the difference between these two directional components, $\text{NEJC}^g_i(H) = C^g_{i\rightarrow\bullet}(H) - C^g_{i\leftarrow\bullet}(H)$, and reflects the extent to which variable i acts as a net transmitter or a net receiver of shocks within the system. Positive values of $\text{NEJC}^g_i(H)$ indicate that variable i contributes more strongly to the propagation of shocks across the network than it absorbs, while negative values imply that it is predominantly influenced by shocks originating elsewhere. By aggregating spillovers across all variables and horizons, the NEJC measure provides a comprehensive characterization of each variable's extended joint contribution to systemic connectedness.

Figure 5: Net Extended Joint Connectedness with Financial Institutions at the 95th Percentile

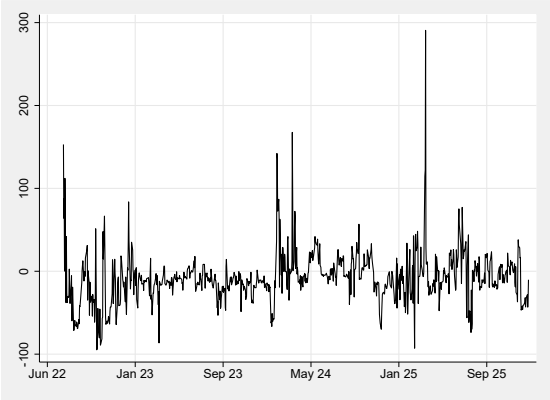
(a) VIX & Banks



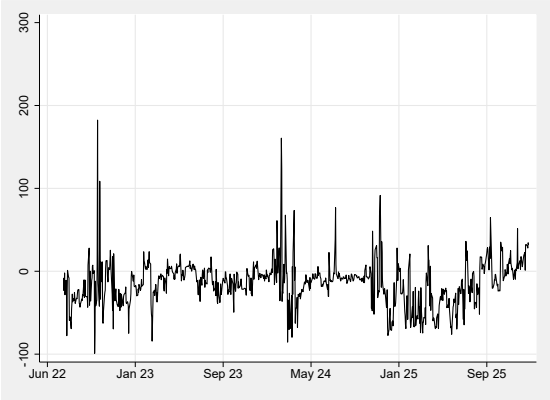
(b) GoldV & Banks



(c) OilV & Banks



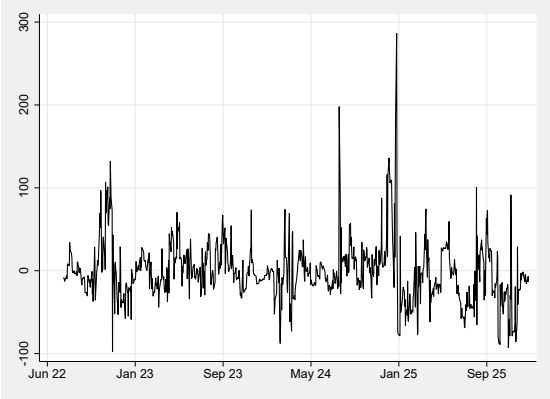
(d) CryptoS & Banks



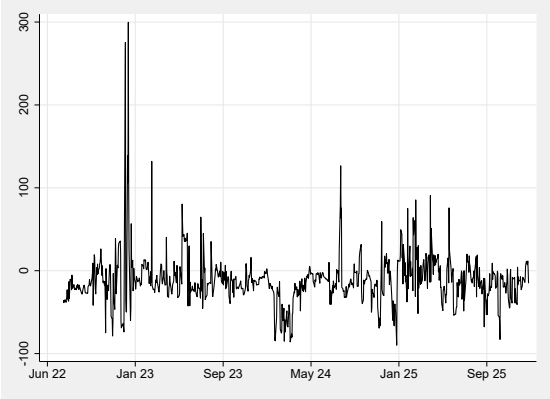
Note: The Net Extended Joint Connectedness (NEJC) measure summarizes the overall systemic role of each variable within the connectedness network by jointly accounting for its transmission and reception of shocks across all horizons considered. Formally, the total directional connectedness *to* others captures the cumulative impact of shocks originating in variable i on all remaining variables j, and is defined as: $C^g_{i\rightarrow\bullet}(H) = \sum_{\substack{j=1\\ j\neq i}}^{k} \psi\widetilde{\psi}^g_{ji}(H)$, where $\psi\widetilde{\psi}^g_{ji}(H)$ denotes the generalized forecast error variance decomposition at horizon H measuring the contribution of shocks in i to the forecast error variance of j. Conversely, the total directional connectedness *from* others measures the aggregate influence of shocks from all remaining variables j on variable i, and is given by $C^g_{i\leftarrow\bullet}(H) = \sum_{\substack{j=1\\ j\neq i}}^{k} \psi\widetilde{\psi}^g_{ij}(H)$. The Net Extended Joint Connectedness for variable i is then defined as the difference between these two directional components, $\text{NEJC}^g_i(H) = C^g_{i\rightarrow\bullet}(H) - C^g_{i\leftarrow\bullet}(H)$, and reflects the extent to which variable i acts as a net transmitter or a net receiver of shocks within the system. Positive values of $\text{NEJC}^g_i(H)$ indicate that variable i contributes more strongly to the propagation of shocks across the network than it absorbs, while negative values imply that it is predominantly influenced by shocks originating elsewhere. By aggregating spillovers across all variables and horizons, the NEJC measure provides a comprehensive characterization of each variable's extended joint contribution to systemic connectedness.

Figure 5: Net Extended Joint Connectedness with Financial Institutions at the 95th Percentile (*continued*)

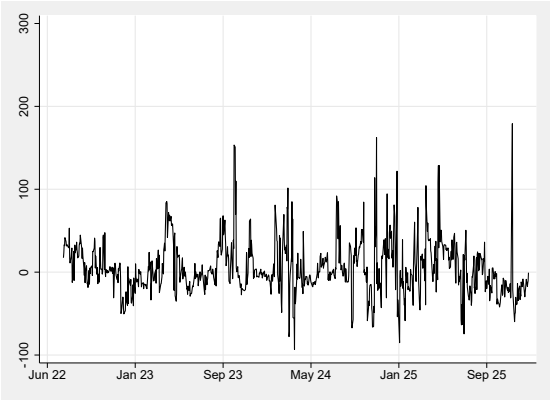
(e) VIX & (GoldV + Banks)



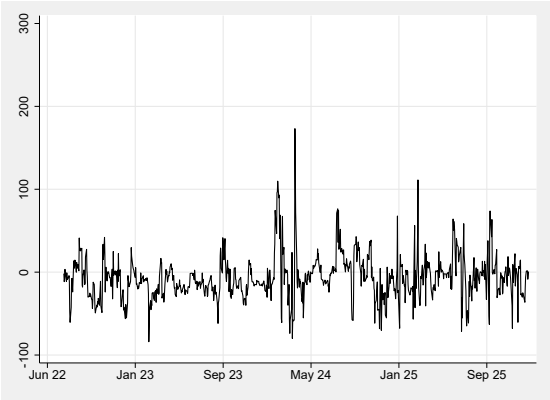
(f) GoldV & (VIX + Banks)



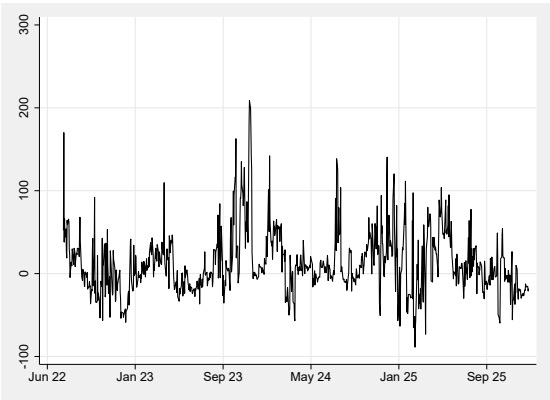
(g) VIX & (OilV + Banks)



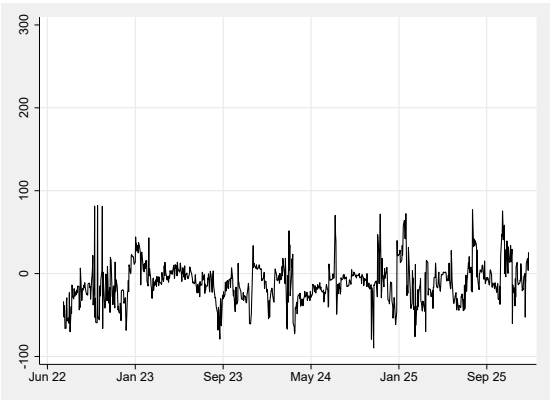
(h) OilV & (VIX + Banks)



(i) VIX & (CryptoS + Banks)

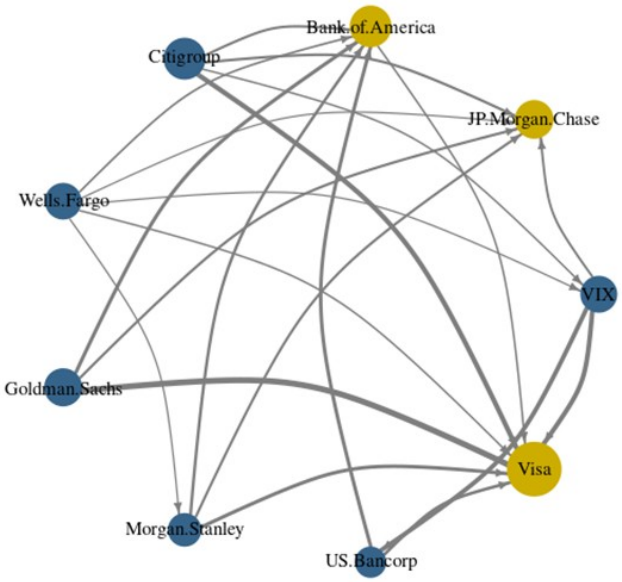


(j) CryptoS & (VIX + Banks)



Note: The Net Extended Joint Connectedness (NEJC) measure summarizes the overall systemic role of each variable within the connectedness network by jointly accounting for its transmission and reception of shocks across all horizons considered. Formally, the total directional connectedness *to* others captures the cumulative impact of shocks originating in variable i on all remaining variables j, and is defined as: $C^g_{i\rightarrow\bullet}(H) = \sum_{\substack{j=1\\ j\neq i}}^{k} \psi\tilde{\psi}^g_{ji}(H)$, where $\psi\tilde{\psi}^g_{ji}(H)$ denotes the generalized forecast error variance decomposition at horizon H measuring the contribution of shocks in i to the forecast error variance of j. Conversely, the total directional connectedness *from* others measures the aggregate influence of shocks from all remaining variables j on variable i, and is given by $C^g_{i\leftarrow\bullet}(H) = \sum_{\substack{j=1\\ j\neq i}}^{k} \psi\tilde{\psi}^g_{ij}(H)$. The Net Extended Joint Connectedness for variable i is then defined as the difference between these two directional components, $\text{NEJC}^g_i(H) = C^g_{i\rightarrow\bullet}(H) - C^g_{i\leftarrow\bullet}(H)$, and reflects the extent to which variable i acts as a net transmitter or a net receiver of shocks within the system. Positive values of $\text{NEJC}^g_i(H)$ indicate that variable i contributes more strongly to the propagation of shocks across the network than it absorbs, while negative values imply that it is predominantly influenced by shocks originating elsewhere. By aggregating spillovers across all variables and horizons, the NEJC measure provides a comprehensive characterization of each variable's extended joint contribution to systemic connectedness.

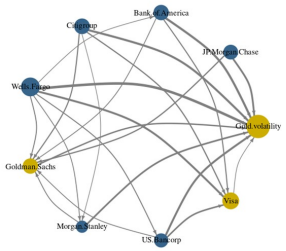
Figure 6: Network connectedness of VIX and major banks at the 5th quantile



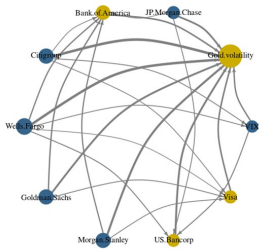
Note: The above network analyses are based on the total connectedness which measures the joint impact that is generated from variable i to all the remaining variables j: $C^g_{i\to j}(H) = \sum_{j=1,i\neq j}^{k} \tilde{\psi}^g_{ji}(H)$. Moreover, the influential impact of shocks from all the remaining variables j on variable i is illustrated by the total directional connectedness FROM others: $C^g_{i\leftarrow j}(H) = \sum_{j=1,i\neq j}^{k} \tilde{\psi}^g_{ji}(H)$. The net effect that variable i produces on the network investigated is the difference between the total directional connectedness TO others and the total directional connectedness FROM others. This leads to the net total directional connectedness. The blue node denotes that this variable is a net source of spillover effects, whereas the yellow node indicates that this variable is a net receiver of spillover impacts. The node size indicates the strength of this variable's net source or reception. Arrows depict the direction of spillover causality.

Figure 7: Network connectedness between selected assets at the 5th quantile

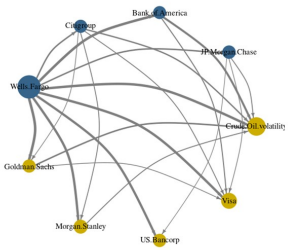
(a) Gold Volatility & Major Banks



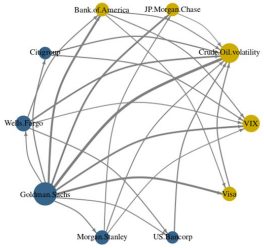
(b) Oil Volatility & Major Banks



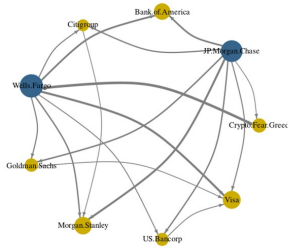
(c) Crypto Sentiment & Major Banks



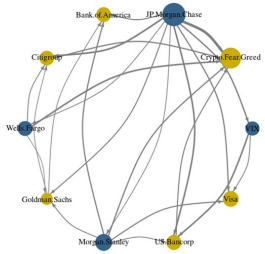
(d) Gold Volatility w/VIX & Major Banks



(e) Oil Volatility w/VIX & Major Banks

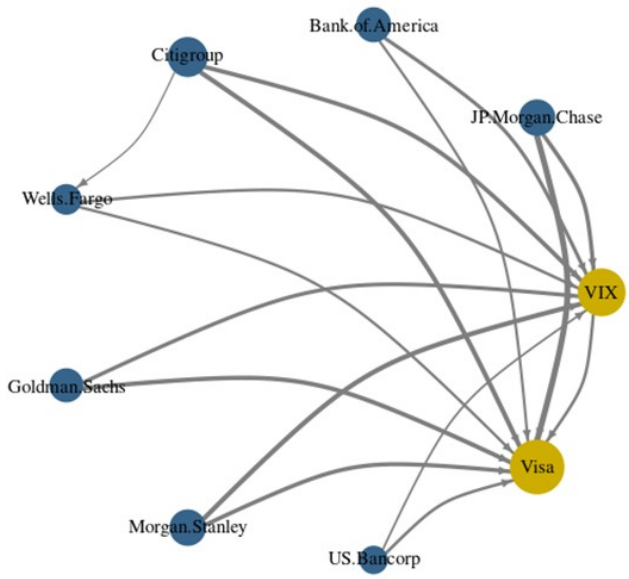


(f) Crypto Sentiment w/VIX & Major Banks



Note: The above network analyses are based on the total connectedness which measures the joint impact that is generated from variable i to all the remaining variables j: $C^g_{i\to j}(H) = \sum_{j=1, i\neq j}^{k} \tilde{\psi}^g_{ji}(H)$. Moreover, the influential impact of shocks from all the remaining variables j on variable i is illustrated by the total directional connectedness FROM others: $C^g_{i\leftarrow j}(H) = \sum_{j=1, i\neq j}^{k} \tilde{\psi}^g_{ji}(H)$. The net effect that variable i produces on the network investigated is the difference between the total directional connectedness TO others and the total directional connectedness FROM others. This leads to the net total directional connectedness. The blue node denotes that this variable is a net source of spillover effects, whereas the yellow node indicates that this variable is a net receiver of spillover impacts. The node size indicates the strength of this variable's net source or reception. Arrows depict the direction of spillover causality.

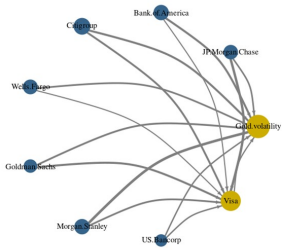
Figure 8: Network connectedness of VIX and major banks at the 50th quantile



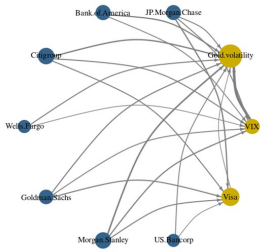
Note: The above network analyses are based on the total connectedness which measures the joint impact that is generated from variable i to all the remaining variables j: $C^g_{i\to j}(H) = \sum_{j=1,i\neq j}^{k} \tilde{\psi}^g_{ji}(H)$. Moreover, the influential impact of shocks from all the remaining variables j on variable i is illustrated by the total directional connectedness FROM others: $C^g_{i\leftarrow j}(H) = \sum_{j=1,i\neq j}^{k} \tilde{\psi}^g_{ji}(H)$. The net effect that variable i produces on the network investigated is the difference between the total directional connectedness TO others and the total directional connectedness FROM others. This leads to the net total directional connectedness. The blue node denotes that this variable is a net source of spillover effects, whereas the yellow node indicates that this variable is a net receiver of spillover impacts. The node size indicates the strength of this variable's net source or reception. Arrows depict the direction of spillover causality.

Figure 9: Network connectedness between selected assets at the 50th quantile

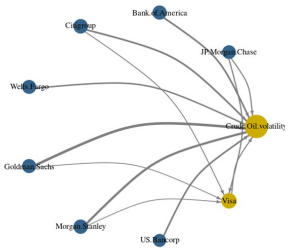
(a) Gold Volatility & Major Banks



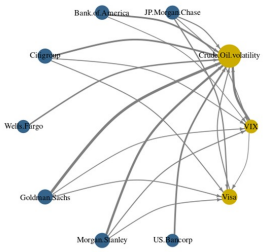
(b) Oil Volatility & Major Banks



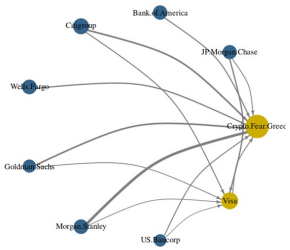
(c) Crypto Sentiment & Major Banks



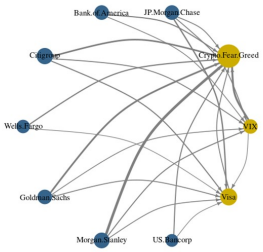
(d) Gold Volatility w/VIX & Major Banks



(e) Oil Volatility w/VIX & Major Banks

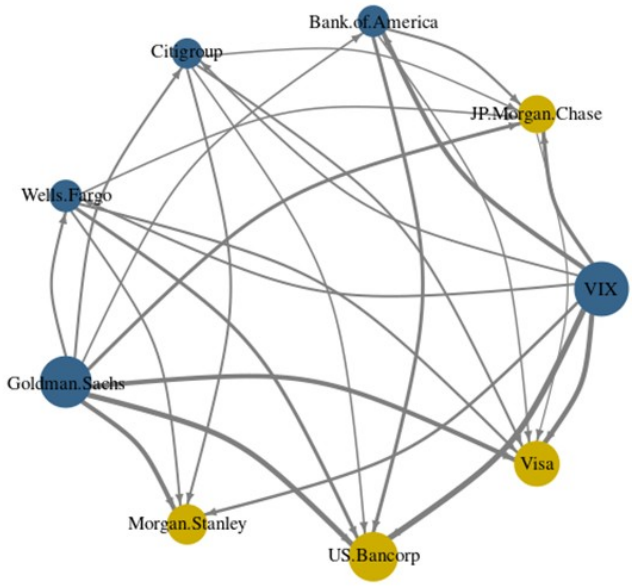


(f) Crypto Sentiment w/VIX & Major Banks



Note: The above network analyses are based on the total connectedness which measures the joint impact that is generated from variable i to all the remaining variables j: $C^g_{i\to j}(H) = \sum_{j=1, i\neq j}^{k} \tilde{\psi}^g_{ji}(H)$. Moreover, the influential impact of shocks from all the remaining variables j on variable i is illustrated by the total directional connectedness FROM others: $C^g_{i\leftarrow j}(H) = \sum_{j=1, i\neq j}^{k} \tilde{\psi}^g_{ji}(H)$. The net effect that variable i produces on the network investigated is the difference between the total directional connectedness TO others and the total directional connectedness FROM others. This leads to the net total directional connectedness. The blue node denotes that this variable is a net source of spillover effects, whereas the yellow node indicates that this variable is a net receiver of spillover impacts. The node size indicates the strength of this variable's net source or reception. Arrows depict the direction of spillover causality.

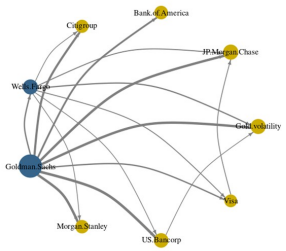
Figure 10: Network connectedness of VIX and major banks at the 95th quantile



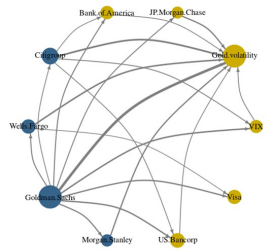
Note: The above network analyses are based on the total connectedness which measures the joint impact that is generated from variable i to all the remaining variables j: $C^g_{i\to j}(H) = \sum_{j=1,i\neq j}^{k} \tilde{\psi}^g_{ji}(H)$. Moreover, the influential impact of shocks from all the remaining variables j on variable i is illustrated by the total directional connectedness FROM others: $C^g_{i\leftarrow j}(H) = \sum_{j=1,i\neq j}^{k} \tilde{\psi}^g_{ji}(H)$. The net effect that variable i produces on the network investigated is the difference between the total directional connectedness TO others and the total directional connectedness FROM others. This leads to the net total directional connectedness. The blue node denotes that this variable is a net source of spillover effects, whereas the yellow node indicates that this variable is a net receiver of spillover impacts. The node size indicates the strength of this variable's net source or reception. Arrows depict the direction of spillover causality.

Figure 11: Network connectedness between selected assets at the 95th quantile

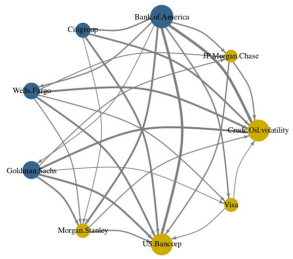
(a) Gold Volatility & Major Banks



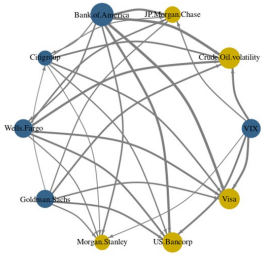
(b) Oil Volatility & Major Banks



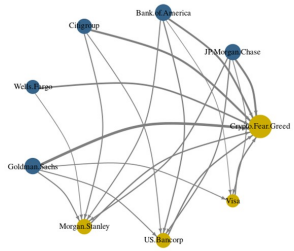
(c) Crypto Sentiment & Major Banks



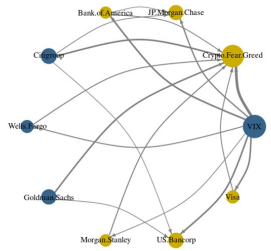
(d) Gold Volatility w/VIX & Major Banks



(e) Oil Volatility w/VIX & Major Banks



(f) Crypto Sentiment w/VIX & Major Banks



Note: The above network analyses are based on the total connectedness which measures the joint impact that is generated from variable i to all the remaining variables j: $C^g_{i\to j}(H) = \sum_{j=1, i\neq j}^{k} \tilde{\psi}^g_{ji}(H)$. Moreover, the influential impact of shocks from all the remaining variables j on variable i is illustrated by the total directional connectedness FROM others: $C^g_{i\leftarrow j}(H) = \sum_{j=1, i\neq j}^{k} \tilde{\psi}^g_{ji}(H)$. The net effect that variable i produces on the network investigated is the difference between the total directional connectedness TO others and the total directional connectedness FROM others. This leads to the net total directional connectedness. The blue node denotes that this variable is a net source of spillover effects, whereas the yellow node indicates that this variable is a net receiver of spillover impacts. The node size indicates the strength of this variable's net source or reception. Arrows depict the direction of spillover causality.